Master in Business Analytics & Data Science
Research Capstone in Operations Research

The Price of Sophistication: When Do Spatial and Robust Modeling Pay in Carbon-Aware Data-Center Scheduling?

A Day-Ahead Migration Scheduler and a Complexity–Value Decision Rule

Extended version (transfer channel and validation chapter in the main body)

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AI Use Declaration. Generative AI tools were used as a research and engineering assistant under the author's direction: for code implementation support, experiment orchestration, literature triage, and drafting assistance. All research questions, modelling decisions, experimental designs, and interpretations are the author's, developed with the supervisor; all results were produced by version-controlled code reviewed by the author (with the Phase 1–2 falsification protocol pre-registered; the Part 3 transfer and emergency-crossover sweeps are exploratory extensions, not part of that pre-registration), and all text was reviewed and edited by the author.

Abstract

Data-center operators can shift compute toward cleaner hours and regions, and recent work applies distributionally robust optimization (DRO) to carbon-aware scheduling. The unexamined question is which modelling layers actually pay for their complexity. This thesis builds a working day-ahead carbon-aware migration scheduler and characterises the value of each layer. Over the honest baseline (carbon-aware scheduling with no inter-region transfer, $\Phi = 0$, on the same feasible set), active migration cuts daily $\mathrm{CVaR}_{0.95}$ emissions by 4.0–9.9% (Western 4.0%, Eastern 9.9%, Diversified 9.0%), and this deterministic transfer channel is the dominant lever; the carbon-computing literature independently finds spatial shifting dominates temporal shifting. This thesis introduces a pre-registered, falsification-based methodology for *pricing* each modelling layer, separating the *validity* of the spatial-dependence assumption (the cross-region carbon correlations are real) from its *value* to the decision (a replicated null). We then isolate the value of *passive* sophistication. Using a Mahalanobis–Wasserstein DRO solved as a second-order cone program, a pre-registered shuffled-marginals falsification test across three real US/Canada grids spanning the dependence spectrum (the US West, an Ontario-anchored Eastern belt, and an engineered solar/wind/hydro portfolio), with an Iberia–France anchor, returns a replicated **null**: encoding cross-region covariance adds below 0.4% of CVaR, surviving Ledoit–Wolf shrinkage, removing each region's average daily pattern (residualization), Benjamini–Hochberg correction, walk-forward validation, and a per-cell equivalence test. A mean-leveling test shows the covariance signal is genuine but masked by the mean field (a mean-dominance bound), and a tail-dependence analysis shows the residual dependence is non-elliptical and upper-tail independent, hence invisible to covariance-based ambiguity sets; a copula appendix confirms even the maximal coupling recovers nothing. Finally we ask when robustness itself pays. Following the price of robustness of Bertsimas and Sim, a DRO over day-ahead forecast error buys a small worst-day (CVaR) tail reduction at a mean premium, with a value of the stochastic solution near zero, and pays over deterministic transfer only past an emergency-severity crossover $M^\star \approx 3$ (with M the multiple by which a region's carbon intensity spikes in a rare grid emergency, so $M = 3$ is a tripling); tested against real data-grounded worst-tail emergencies across 17 zones, that crossover never activates. This extended version develops the active transfer channel and its severity crossover directly in the main body (Section 6), takes the controller *online* as a rolling-horizon problem under forecast error (Section 7), and generalises the mean-dominance bound into a dimensionless problem-class condition (Section 8); a dedicated chapter (Section 9) documents the validation behind every claim: small-instance model checks, a pre-registered protocol, and a 199-test software suite. The thesis contributes a working scheduler with honest savings, a screening rule for when passive dependence modelling fails, and a price-of-robustness decision rule with a tested, data-grounded bound, all under 199 unit tests, with the Phase 1–2 falsification protocol pre-registered (the stylised-emergency crossover of RQ3 is a calibrated stress test, not part of the

pre-registration).

Contents

1 Introduction

1.1 Context

Data centers consumed an estimated 1–2% of global electricity and their demand is growing rapidly with AI workloads [10]. Unlike most industrial loads, compute is unusually *flexible*: many batch and training workloads can be delayed by hours or moved between sites with little loss of utility. This movement is *logical*, jobs relocate over the network between an operator's data centers, and does not require the regions' power grids to be physically interconnected; cross-region correlation of carbon intensity therefore matters as information for the scheduler, not as an electrical constraint.[1] Because the carbon intensity of grid electricity varies strongly by hour and by region, driven by wind, solar, hydro availability, and demand, this flexibility lets a data-center operator act as a grid-aware participant, deferring work from dirty hours to clean ones [20, 23]. Google's carbon-intelligent computing platform demonstrated such temporal shifting at production scale [20], and recent systems work extends it to variable-capacity operation [24].

Scheduling against *tomorrow's* carbon intensity, however, means scheduling under uncertainty. Distributionally robust optimization (DRO) addresses this by optimizing against the worst distribution in an ambiguity ball around the empirical distribution [8], and has been applied to carbon-aware scheduling with carbon intensity as the uncertain quantity [9]. Existing formulations treat each region's carbon intensity essentially in isolation. Yet carbon intensities of electrically interconnected regions co-move: weather fronts traverse neighbouring balancing areas, and shared solar fleets create synchronized clean periods. It is therefore natural to model carbon intensity as a stochastic *vector* with a full cross-region covariance, the extension studied here.

1.2 Problem statement and research questions

Each modelling layer, cross-region covariance, distributional robustness, and richer copulas, adds estimation and computational cost, and is worthwhile only if it buys measurably better schedules. The literature has assumed, rather than measured, that this sophistication pays. This thesis builds the scheduler and measures the value of each layer directly.

RQ1. How much can a day-ahead carbon-aware migration scheduler cut daily $\mathrm{CVaR}_{0.95}$ emissions relative to carbon-aware scheduling without inter-region transfer, and which lever drives the saving?

[1] We use "spatial" throughout in the sense of *across the operator's geographically separate regions* (sites), as in the standard spatial-versus-temporal load-shifting distinction. The dependence we study is the cross-region statistical correlation of carbon intensity, driven by shared weather and generation; it is not distance-decay spatial autocorrelation (e.g. Moran's I or a variogram) and not an electrical coupling of the grids, which need not be interconnected.

RQ2. When does adding passive modelling complexity (cross-region covariance, or a non-elliptical copula) improve out-of-sample tail emissions, and at what cost?

RQ3. When does robustifying against day-ahead forecast error pay over deterministic transfer, and do real grids ever reach that regime? What decision rule follows?

1.3 Contributions

The thesis makes five contributions. (i) *A working day-ahead scheduler with honest, baseline-anchored savings*, developed in the main body as an active transfer channel (Section 6): active inter-region migration cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% over the no-transfer ($\Phi = 0$) baseline on the same feasible set across three real grids, via a spatially-coupled transfer DRO that makes inter-region flows a decision. The transfer mechanism follows established carbon-aware practice; the contribution is the $\Phi = 0$-anchored measurement of its spatial value. (ii) *A pre-registered methodology for pricing modelling layers*: holding the optimization fixed and manipulating one object at a time, it measures the marginal out-of-sample CVaR value of each layer, and separates the *validity* of the spatial-dependence assumption (correlations are genuine, 0.7–0.8) from its *value* to the decision (null). To our knowledge this is the first treatment to place carbon intensity itself in a Mahalanobis–Wasserstein ball as a correlated cross-region vector and isolate the marginal value of that coupling via a shuffled-marginals falsification test. (iii) *A screening rule* for when passive dependence modelling cannot help, supported from three independent directions: an a-priori SOCP gap-bound (Proposition 1) before any data is touched; a mean-leveling test showing the covariance signal is genuine but masked by the mean field; and a tail-dependence/comonotone-copula argument showing even the maximal coupling recovers nothing. A dimensionless condition Δ (Section 8) generalises the screen beyond carbon, reported as a coarse ordinal indicator (it does not bind on the real grids, so the empirical falsification remains necessary there). (iv) *A price-of-robustness decision rule* with a data-grounded, falsified crossover, taken online as a rolling-horizon controller (Section 7): robustification pays only past an emergency-severity crossover $M^\star \approx 3$ that realized severity (1.3–1.9×; Winter Storm Uri 1.3×) never reaches, stated as the rule plus its tested bound, not a universal impossibility. (v) *A reproducible, validated pipeline* (Section 9): pre-registration, 199 unit tests, a small-instance model validation, a robustness battery, Benjamini–Hochberg correction, and per-cell TOST equivalence tests that affirmatively reject any effect above the 0.4% materiality margin.

1.4 Thesis structure

Section 2 reviews the carbon-aware scheduling and DRO literature and locates the gap: the value of modelling sophistication has been assumed, not measured. Section 3 develops the day-ahead migration scheduler, the DRO model, the falsification test, and the evaluation protocol. Section 4 reports the scheduler savings and the value of each modelling layer: where the value comes from, the screening rule for passive dependence, and the price-of-robustness decision rule with the severity crossover. Section 5 interprets the findings against prior work and states limitations. Section 6 then develops the active transfer channel in full, the spatially-coupled transfer DRO and its severity crossover, with a data-grounded emergency stress test; Section 7 takes it *online* as a rolling-horizon controller; and Section 8 generalises the mean-dominance bound into a problem-class condition. Section 9 consolidates the model, protocol, and software validation behind every claim. Section 10 concludes with the practitioner decision rule and future work, and Appendix C records the five-part programme's status and horizon.

1.5 A note on terminology

For readability the main technical terms are glossed in plain language here; the precise definitions follow in Section 3.

Carbon intensity: how dirty the electricity is at a given hour and place (grams of CO_2 per kWh); it swings with how much wind, solar, and hydro are on the grid.

Distributionally robust optimization (DRO): planning against the worst-case carbon distribution within a set of plausible distributions around the data, rather than trusting a single point forecast. Like packing for weather a little worse than the forecast.

Wasserstein distance / ball: a way to measure how far apart two probability distributions are (the cheapest cost of reshaping one into the other). The Wasserstein *ball* of radius ε is the set of all distributions within that distance of the data, every future close enough to what we observed to be worth hedging.

$\mathrm{CVaR}_{0.95}$ **(Conditional Value-at-Risk):** the average emissions on the worst 5% of days; a tail-risk measure, not the average day.

Second-order cone program (SOCP): the convex problem the scheduler reduces to, solved to global optimality in a fraction of a second.

Mahalanobis distance: a distance that weights directions by the carbon covariance, so it accounts for how regions move together. Like measuring distance on a stretched map where correlated regions sit closer.

Copula: the part of a joint distribution that captures only how regions move together, with each region's own behaviour stripped out.

Comonotone: the strongest possible copula, all regions rising and falling in perfect lockstep; used as a worst-case upper bound.

Tail dependence ($\chi_{\mathrm{U}}, \chi_{\mathrm{L}}$): whether regions hit their dirtiest (χ_{U}) or cleanest (χ_{L}) hours at the same time, a joint-extreme effect plain correlation can miss.

Residualization: removing each region's predictable hour-of-day (or day-to-day) average so the analysis works only on the genuine surprises, not the routine daily cycle.

Shuffled-marginals test: re-fitting the scheduler with the cross-region link destroyed but each region's own pattern intact, so the difference isolates the value of cross-region correlation.

Mean-leveling test: we deliberately flatten each region's average carbon level (so the only signal left is the cross-region co-movement) and re-run the scheduler. It shows whether that cross-region structure is genuinely useful but normally hidden behind the much larger differences in average carbon.

Mean-dominance (Proposition 1): an a-priori bound showing the average carbon field caps how much any dependence model can change the schedule.

Out-of-sample / walk-forward: scoring a plan only on data it never trained on (out-of-sample); walk-forward is the rolling version, train on earlier years and test on a later untouched one, mimicking real deployment.

Equivalence test (TOST): a test that affirmatively shows an effect is smaller than a chosen materiality margin, turning "we found nothing" into "any effect this large is ruled out."

Severity M: the multiple by which one region's carbon spikes in a rare grid emergency ($M = 3$ is a tripling); the crossover $M^\star$ is where robustness starts to pay.

Value of the stochastic solution (VSS): how much planning against the full uncertainty distribution beats committing to a single point forecast; here it is near zero.

Transfer budget (Φ): how much compute may migrate between regions; $\Phi = 0$ is the honest no-transfer baseline, and raising it delivers the headline savings.

2 Literature Review

2.1 Carbon-aware computing

Carbon-aware scheduling exploits temporal and spatial variation in grid carbon intensity. In production at Google, Radovanović et al. [20] describe a system which computes day-ahead "virtual capacity curves" that throttle flexible compute during high-carbon hours; measured impact on latency-sensitive workloads is negligible while load shifts measurably toward cleaner hours. Wiesner et al. [23] analyse the potential of temporal shifting across regions and workloads, finding the benefit depends strongly on the grid's marginal mix and the workload's flexibility. Wijayawardana and Chien [24] study cloud-VM scheduling on datacenters whose capacity varies with grid conditions (including carbon-coupled capacity, where a constant carbon budget forces capacity to fall as intensity rises) and document the goodput cost of capacity variation for long-running jobs. This systems literature establishes operational levers (deferral, throttling, capacity coupling) that we adopt as constraints, but it treats future carbon intensity as a point forecast; uncertainty is handled implicitly, by re-planning. That geographic migration itself lowers carbon, moving compute toward whichever region is cleaner, is established by this line of work [20, 23]; we take the mechanism as given. What remains assumed rather than measured is whether *modelling* the joint spatial structure, or hedging forecast error robustly, adds anything on top, the modelling layers this thesis prices.

2.2 Distributionally robust optimization and the price of robustness

DRO optimizes against the worst-case distribution within an ambiguity set, and the Wasserstein ball around the empirical distribution has emerged as the canonical data-driven choice, with tractable finite-dimensional reformulations [8, 18]. For costs linear in the uncertain vector, type-2 Wasserstein DRO with a Mahalanobis ground metric reduces to a mean term plus a covariance-weighted regularizer, an interpretable "mean plus $\varepsilon \sqrt{x^\top \Sigma x}$" objective solvable as a second-order cone program (SOCP) [2]. Hall et al. [9] apply Wasserstein DRO to carbon-aware data-center scheduling with virtual capacity curves, treating each region's carbon intensity as the uncertain quantity and demonstrating robustness gains over sample-average approximation. The risk functional we target, CVaR_β, is standard for tail-focused evaluation [21]. Beyond second moments, recent work robustifies the *dependence structure* itself: Fan et al. [5] place both the marginals and the *copula* within a Wasserstein ball, the natural home for the non-elliptical structure our copula schedulers target.

Robustness is never free. Bertsimas and Sim [3] framed this as the *price of robustness*: protecting against worst-case realizations trades a guaranteed degradation in nominal performance for a bounded improvement in the tail, and the operator chooses where on that frontier to sit. This lens is central to our RQ3: we do not ask whether robustification *can* help (it can, in the tail) but whether the price it charges is paid back under conditions real grids actually reach. The mean-premium-for-tail-reduction trade-off we report for the day-ahead scheduler is a direct instance of their frontier.

2.3 Modelling cross-region dependence

Cross-region dependence enters a Mahalanobis–Wasserstein model only through the off-diagonal blocks of the covariance matrix. Estimating large covariance matrices from short panels is noisy; shrinkage estimators [19] are the standard remedy and serve here as a robustness arm. Beyond second moments, dependence is fully described by the copula [11, 22]; tail-dependence coefficients quantify joint extremes that correlation cannot see, and the Gaussian copula has zero asymptotic tail dependence for any correlation below one [7], a property we use as a diagnostic benchmark. Vine (pair-copula) constructions make high-dimensional non-elliptical dependence tractable [11, 12, 14], with established sequential structure-selection procedures [13], and copula-based scenario generation for stochastic programs is well developed [15]. Whether richer non-elliptical structure (vine copulas, or copula-ambiguity DRO [5]) improves a *stochastic program* depends on whether it changes the optimal decision out-of-sample; we test this directly and report it in Appendix B.

2.4 The gap: value has been assumed, not measured

The carbon-aware scheduling literature establishes that *temporal* shifting (deferral to clean hours) delivers measurable savings, confirmed operationally at Google and in recent studies [20, 23]. The natural extension is *spatial* shifting (migration toward clean regions), and the distributed-computing literature finds spatial shifting dominates temporal shifting in potential [23], independently motivating the transfer channel studied here. Yet the value of *modelling* that spatial structure (via cross-region covariance, copulas, or distributional robustness) has been *assumed*, not measured. The DRO literature applies sophisticated ambiguity sets to uncertain vectors without isolating the marginal value of the cross-coordinate dependence they encode, and neither literature offers a controlled mechanism for *why* spatial structure should or should not matter. Adjacent work finds simple carbon-aware heuristics capture most attainable savings, with added sophistication offering little incremental benefit [4], and the role of dependence structure in Wasserstein DRO has been studied abstractly [6]; our wedge is the controlled, pre-registered, triangulated *measurement* of the dependence layer's value, not the qualitative conclusion. This thesis fills that gap: we build a working day-ahead scheduler that actively migrates compute and measure the savings (RQ1), isolate the value of passive modelling sophistication via a falsification test with a mechanistic explanation (RQ2), and frame robustification as a price-of-robustness trade-off, testing when that price is paid in real data (RQ3).

3 Methodology

3.1 Research design overview

The design tests whether passive (covariance-based) spatial modelling of carbon intensity adds value to robust day-ahead scheduling, and measures the value of active transfer separately. We hold the optimization machinery fixed and manipulate exactly one object (the cross-region blocks of the covariance supplied to the scheduler) then measure the consequence on held-out tail emissions. Three primary region sets spanning the dependence spectrum (with two earlier panels corroborating) serve as replications; three nested constraint regimes guard against the result being an artifact of a particular feasible-set geometry; a robustness battery (estimators, multiple-testing correction, walk-forward) guards against statistical artifacts; and two further experiments (mean-leveling test; tail dependence) identify the mechanism. Throughout, a pre-registration discipline removes analyst degrees of freedom.

3.2 The scheduling problem

Compute is scheduled across R regions over a $T = 24$-hour day. The decision is $x \in \mathbb{R}_{\geq 0}^{R\times T}$, where $x_{r,t}$ is compute power (MW) in region r at hour t. Carbon intensity $\rho_{r,t}$ (gCO$_2$eq/kWh) is uncertain; realized emissions are linear, $\langle \rho, x\rangle = \sum_{r,t} \rho_{r,t} x_{r,t}$.

3.3 Distributionally robust formulation: day-ahead forecast-error hedging

The robust layer hedges *day-ahead forecast error*. Each day the scheduler commits against a day-ahead point forecast $\bar{\rho}$ of the carbon field (the hour-of-day climatological mean on the training window, updated daily); the realized field departs by a residual $\xi = \rho - \bar{\rho}$ whose empirical distribution $\hat{\mathbb{P}}$, pooled over 2021–2024, defines the ambiguity. We minimize worst-case expected emissions over a type-2 Wasserstein ball $\mathcal{B}_\varepsilon(\hat{\mathbb{P}})$ of radius ε centred on that residual distribution, with Mahalanobis ground metric induced by the empirical residual covariance $\hat{\Sigma} \in \mathbb{R}^{RT\times RT}$:

$$\min_{x\in\mathcal{X}} \sup_{\mathbb{Q}\in\mathcal{B}_\varepsilon(\hat{\mathbb{P}})} \mathbb{E}_{\mathbb{Q}}\big[\langle \rho, x\rangle\big] \;=\; \min_{x\in\mathcal{X}} \langle \bar{\rho}, x\rangle + \varepsilon\sqrt{x^\top \hat{\Sigma}\, x}, \tag{1}$$

by Wasserstein duality for linear costs [2, 8], where $\bar{\rho}$ is the empirical mean field. With the Cholesky factorization $\hat{\Sigma} = LL^\top$ the penalty becomes $\varepsilon\|L^\top \operatorname{vec}(x)\|_2$. Introducing an epigraph variable η (distinct from the hour index t), (1) is the second-order cone program

$$\min_{x\in\mathcal{X},\, \eta\geq 0} \langle \bar{\rho}, x\rangle + \varepsilon\,\eta \quad \text{s.t.} \quad \big\|L^\top \operatorname{vec}(x)\big\|_2 \leq \eta, \tag{2}$$

solved to global optimality (CLARABEL). The radius ε interpolates from the deterministic LP ($\varepsilon = 0$) to increasingly conservative schedules.

Where cross-region correlation enters. Partitioning $\hat{\Sigma}$ into $T \times T$ blocks Σ_{rs} by region pair, the quadratic form splits as

$$x^\top \hat{\Sigma} x = \sum_r x_r^\top \Sigma_{rr} x_r + \sum_{r \neq s} x_r^\top \Sigma_{rs} x_s, \tag{3}$$

so the off-diagonal (spatial) blocks are the *only* channel by which cross-region dependence can alter the schedule. The experiment manipulates exactly these blocks.

3.4 A ladder of modelling layers

The thesis studies schedulers arranged by sophistication: **(0)** carbon-blind; **(1)** per-region deterministic (each region by its own mean forecast); **(2)** per-region robust (single-region Wasserstein DRO, no cross-region structure); **(3)** joint covariance (the SOCP (1) with the full spatial Σ, the passive spatial extension); and **(4)** joint copula (non-elliptical dependence; Appendix B). The screening test (Section 3.7) measures the marginal value of (3) over (2). Active transfer ($\Phi > 0$, inter-region flows) is orthogonal to this passive hierarchy and is the dominant lever, studied in Sections 4.2, 4.9 and 6.

3.5 Feasible set and constraint regimes

The feasible set $\mathcal{X}$ collects the linear constraints below; introducing an auxiliary flexible-load variable $x^{\mathrm{f}} \in \mathbb{R}_{\geq 0}^{R \times T}$, the scheduler solves (1) over

$$
\begin{aligned}
&0 \leq x_{r,t} \leq \bar{x}_{r,t}, && \forall r, t && \text{(C0: capacity)} && \text{(4a)}\\
&x_{r,t} = \alpha_r W_r \, p_{r,t} + x^{\mathrm{f}}_{r,t}, && \forall r, t && \text{(C2a: flex/inflex split)} && \text{(4b)}\\
&\textstyle\sum_t x^{\mathrm{f}}_{r,t} = (1 - \alpha_r)\, W_r, && \forall r && \text{(C2b: work conservation)} && \text{(4c)}\\
&|x_{r,t} - x_{r,t-1}| \leq \Delta_r, && \forall r, t && \text{(C3: ramp)} && \text{(4d)}\\
&\textstyle\sum_{t \in \mathcal{W}} x^{\mathrm{f}}_{r,t} \geq \gamma\, (1 - \alpha_r)\, W_r, && \forall r && \text{(3a: deferral deadline)} && \text{(4e)}\\
&\pi_{r,t}\, x_{r,t} \leq \bar{P}, && \forall r, t && \text{(3b: thermal/PUE)} && \text{(4f)}\\
&\textstyle\sum_{r,t} \bar{\rho}_{r,t}\, x_{r,t} \leq B && && \text{(3d: carbon budget)} && \text{(4g)}
\end{aligned}
$$

where W_r is region r's daily work (utilization fixed at 80%, $W_r = 960$ MWh); $\alpha_r \in \{0.30, 0.50, 0.75\}$ is the inflexible fraction following the fixed intraday shape $p_{r,t}$ ($\sum_t p_{r,t} = 1$); $\Delta_r = 15$ MW/h is the ramp limit; $\mathcal{W} = [0, 7]$ and $\gamma = 0.20$ set the deferral deadline; and $\pi_{r,t} = \mathrm{PUE}_0 + \kappa \max(\mathcal{T}_{r,t} - \mathcal{T}_{\mathrm{set}}, 0)$

is the temperature-coupled power-usage multiplier (data, hence a constant per cell), capped at effective power $\bar{P}$. Every constraint is linear, so $\mathcal{X}$ is polyhedral and the program (1) remains an SOCP. (We follow the source labelling: C0–C3 are the base operational constraints and the **3**· series is adapted from the SoCC'25 formulation [24]; a further aggregate hourly cap $\sum_r x_{r,t} \leq p_{\max}$ is implemented but inactive in the reported regimes.)

The capacity ceiling is constant ($\bar{x}_{r,t} = 50$ MW) except under constraint **3c**, the carbon-coupled variable capacity adapted from Wijayawardana and Chien [24], which replaces it with

$$\bar{x}_{r,t} = x_{\min} + (x_{\max} - x_{\min})\,\mathrm{CFE}_{r,t}/100, \tag{5}$$

where $\mathrm{CFE}_{r,t}$ is the training-mean carbon-free-energy share, capacity falls as the grid gets dirtier. The bounds $(x_{\min}, x_{\max}) = (50, 75)$ were calibrated on training data to a loosely binding ("Goldilocks") regime: the constraint binds on 28–37% of region-hours, leaving positive slack and capacity margin, so it shapes the schedule without freezing it.

Three nested regimes report results as a function of operational tightness: **R3** (reference) imposes C0 + C2 + C3; **R1** (lean) adds the deferral deadline 3a; and **R2** adds the thermal ceiling 3b (climate cases) or the variable-capacity ceiling 3c (interconnection cases). The carbon budget 3d is available but inactive in the reported regimes. Each regime is solved as a *complete* model with all of its constraints active jointly; the nesting probes the null's robustness to operational tightness, and is not a test of any single constraint's effect in isolation.

3.6 Model validation on a small instance

Before running the falsification experiment we check that the optimization model itself behaves correctly. The full model is validated *as a whole*, with all constraints of (4) active simultaneously, on a small, human-checkable *two-region* instance (two regions, eight hours, 10 MWh per-cell capacity, $W = 30$ MWh each), rather than each constraint in isolation: correctness is a property of the constraints' *interaction*, and a set that every constraint satisfies alone can still be jointly binding in unexpected ways. We use two regions specifically so the *cross-region* interaction is exercised: a shared aggregate hourly cap $\sum_r x_{r,t} \leq P_{\mathrm{agg}}$ couples the regions, which must then compete for the same clean hours. We fix an analyst-anticipated outcome, perturb one input at a time, and verify the solved schedule moves the anticipated way (Figure 1). At baseline the two regions' clean windows overlap, and the aggregate cap forces them to split the contested hours (panel A). Making region 1's clean window dirty drives it to cede those hours to region 2 and spread to its own next-cleanest hours, with each region's work still conserved (panel B). Tightening the shared cap removes joint headroom, so both regions spread further in time rather than co-locating in the clean window (panel C). All three responses match the hand-anticipated solution and conserve work per region

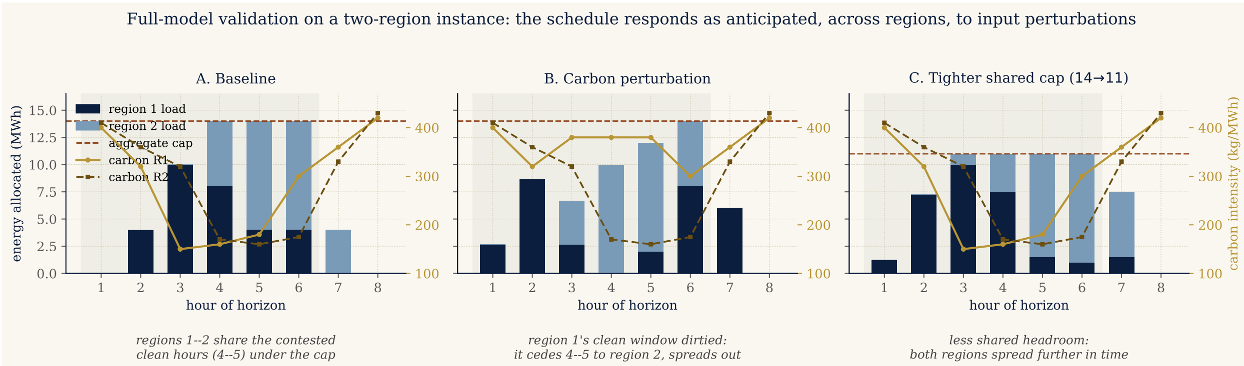

Figure 1: Full-model validation on a two-region instance. The complete feasible set (capacity, work conservation, ramp, deferral deadline) plus a shared aggregate hourly cap is solved jointly; stacked bars show how the two regions share each hour, the dashed line is the aggregate cap, the curves are each region's carbon field, the green band the deadline window. **A:** baseline, the regions split the contested clean hours (4–5) under the cap. **B:** after dirtying region 1's clean window, it cedes those hours to region 2 and spreads out. **C:** after tightening the shared cap, both regions spread further in time. Each response is the one an analyst anticipates, confirming the model encodes the intended cross-region behaviour.

to machine precision. This model-level check is complementary to, and distinct from, the software test suite (Section 3.11, which pins component behaviour such as feasible-set equivalence) and the falsification experiment below (which probes a scientific hypothesis, not model correctness).

3.7 The falsification test: shuffled marginals

The scheduler is fitted twice, with covariances differing in exactly one respect:

$$\left(\hat{\Sigma}^{\mathrm{shuf}}\right)_{rs} = \begin{cases} \Sigma_{rr}, & r = s, \\ \mathbf{0}, & r \neq s, \end{cases} \tag{6}$$

preserving every region's marginal distribution and within-region autocorrelation while destroying only cross-region dependence. Each fitted schedule x is evaluated by the out-of-sample conditional value-at-risk of daily emissions $e_i = \langle \rho_i, x \rangle$ realized on test day i; CVaR is a coherent risk measure [16] standard in stochastic programming [17]. We use the Rockafellar–Uryasev form [21]

$$\mathrm{CVaR}_\beta(e) = \min_{\tau \in \mathbb{R}} \left\{ \tau + \tfrac{1}{1-\beta}\, \mathbb{E}\big[(e-\tau)_+\big] \right\}, \qquad \beta = 0.95, \tag{7}$$

estimated as the empirical mean of the worst 5% of test days. The *spatial gap* is the difference between the two arms,

$$\mathrm{gap} = \mathrm{CVaR}_{0.95}\left(e^{\mathrm{shuf}}\right) - \mathrm{CVaR}_{0.95}\left(e^{\mathrm{joint}}\right), \tag{8}$$

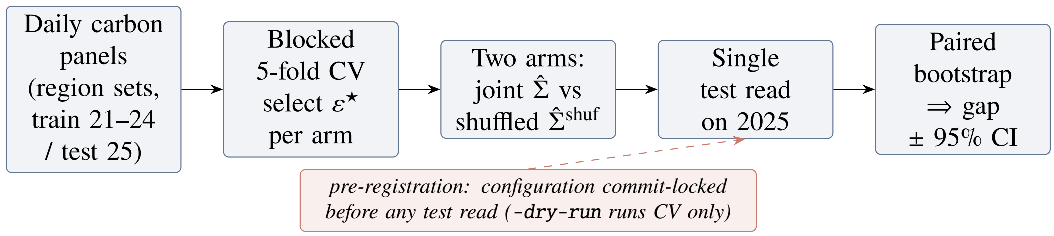

Figure 2: The pre-registered falsification pipeline. Cross-validation and schedule fitting touch only the training years; the 2025 test set is read once, after the configuration is commit-locked, and the spatial gap is read off a paired day-bootstrap.

Algorithm 1 Shuffled-marginals experiment (one region set)

Input: daily panels $\rho_{1:N}$ (train 2021–2024, test 2025); regimes {R3, R1, R2}; α-levels $\{0.30, 0.50, 0.75\}$; grid $\mathcal{E} = \{0, 0.1, 1, 10, 100, 1000\}$
for each regime, each α, each arm $\Sigma \in \{\text{joint}, \text{shuf}\}$ **do**
  for each $\varepsilon \in \mathcal{E}$ **do** ▷ blocked 5-fold time-series CV
    for each contiguous fold k **do**
      fit $\bar{\rho}, L$ on training folds; solve SOCP (1); record validation $\text{CVaR}_{0.95}$
  $\varepsilon^\star \leftarrow \arg\min_\varepsilon$ mean validation CVaR ▷ per arm, independently
  refit on full training set at $\varepsilon^\star$; evaluate *once* on 2025 ▷ single test read
paired bootstrap (1000 resamples, fixed seed) on test days $\Rightarrow$ 95% CI for the gap (8)
Pre-registration: configuration commit-locked before any test read; `-dry-run` gate runs CV only

positive when the joint covariance helps. The test is a screening tool: built to detect a spatial benefit if one exists, it isolates the *value* of passive covariance modelling from the *validity* of cross-region correlation in the data, and validates the mean-dominance bound (Proposition 1). Figure 2 sketches the pipeline and Algorithm 1 gives the full protocol. The recipe is generic: hold the optimizer fixed, perturb exactly one modelling object, pre-register the analysis, and read off that object's marginal out-of-sample tail value; only the perturbed object (here the cross-region covariance) changes when the same protocol is applied to another stochastic program.

3.8 Data

Carbon intensity is hourly lifecycle intensity from Electricity Maps (academic licence), 2021–2025, 43,824 hourly observations per zone; panels are built on a common local clock per region set, trained on 2021–2024 and tested once on 2025. Three primary US/Canada region sets span the dependence spectrum: **(1)** the *US West*, California, Nevada and Phoenix (CISO, BANC, LDWP, NEVP, AZPS); **(2)** an *Ontario–Eastern belt* (CA-ON, NYISO, MISO, PJM); and **(3)** an *engineered heterogeneous portfolio* (CISO (solar), ERCOT (wind), BPAT (hydro), spanning *different* interconnections) deliberately chosen so clean periods do *not* align: the adversarial best case for diversification. This spread is deliberate on both ends of the spectrum. The two correlated, within-interconnection grids are the *best case for the spatial hypothesis*, the most cross-region correlation a covariance model could possibly exploit, and double as a stress test rather than a realistic deployment (no operator clusters data centers within one interconnection). A realistic, globally dispersed fleet looks instead like the heterogeneous portfolio, low and mixed correlation, so the practically relevant case is the one where diversification *should* help most. The null holding at both ends is therefore stronger than a null on either alone. Two earlier panels corroborate: an *Iberia–France* European set (ES, PT, FR), where residual correlation collapses to the diurnal cycle, and the California/Nevada subset of the US West before Phoenix was added at the supervisor's request. Temperature for constraint 3b is hourly ERA5 2 m air temperature (Open-Meteo), one load-weighted point per zone. The licensed raw data is never committed; derived summary tables are archived in the repository for traceability.

3.9 Validation, diagnostics, and the mean-dominance bound

Correlation validation (RQ1). Raw Pearson correlation conflates co-movement with a shared diurnal cycle. We therefore report correlations of *residuals* after removing each zone's hour-of-day mean (local clock), with overlay plots of standardized intensity for summer and winter weeks.

Robustness battery. Every experiment is re-run with (i) Ledoit–Wolf shrinkage covariance [19]; (ii) covariance estimated from seasonal (monthly-mean) residuals; (iii) covariance from AR(1) residuals. A pre-registered agreement rule counts a spatial effect only if positive and sign-stable across both residual estimators. Bootstrap p-values across all cells receive Benjamini–Hochberg correction at $q = 0.05$ [1]; a walk-forward refit (train 2021–2023, test 2024) checks temporal stability; a log-denser ε grid checks grid sensitivity.

The mean-leveling test (RQ2, mechanism). To test whether the mean field masks the covariance, the mean used for *scheduling* is transformed while evaluation stays on real emissions: LEVEL equalizes regional time-averages (keeping hour-of-day shape); FLAT sets a constant mean, making the mean term constant under fixed demand, a covariance-only world in which any joint-vs-shuf difference is attributable purely to the spatial blocks.

Tail dependence (RQ2, structural). For each region pair we estimate the upper and lower tail-dependence coefficients $\chi_{\mathrm{U}}(q) = \Pr(U > q, V > q)/(1 - q)$ and $\chi_{\mathrm{L}}(p) = \Pr(U < p, V < p)/p$ on rank pseudo-observations of residual intensity, comparing χ_{U} at $q = 0.95$ with the Gaussian-copula value at the matched correlation. Since the Gaussian copula, the dependence family a covariance ball can represent, is asymptotically tail-independent [7], a positive empirical excess would be exploitable tail structure invisible to the DRO; and since elliptical copulas force $\chi_{\mathrm{L}} = \chi_{\mathrm{U}}$, radial asymmetry diagnoses non-elliptical dependence.

3.10 Phase 2: richer dependence models (summary)

The covariance ball represents only *elliptical* dependence. To foreclose the "wrong object" objection, Appendix B re-runs the falsification with Gaussian, lower-tail Clayton, and the maximal (comonotone) copula, fitted to the empirical rank structure on the same feasible set. The copula-coupled resampling preserves every region's empirical marginal and varies only the cross-region coupling, exactly as the shuffled covariance does; for each model we draw $S = 1000$ scenarios $\{\rho^s\}$ and solve the sample-average CVaR program in Rockafellar–Uryasev form [21],

$$\min_{x\in\mathcal{X},\,\tau,\,z\geq 0} \tau + \frac{1}{(1-\beta)S}\sum_{s=1}^{S} z_s \quad \text{s.t.} \quad z_s \geq \langle \rho^s, x\rangle - \tau, \tag{9}$$

over the same feasible set $\mathcal{X}$ (4) and regimes as Phase 1. The result confirms the screening rule and is summarized in one paragraph there; Proposition 1 below explains why it must hold.

Why the dependence model is second-order. Two facts make the dependence model second-order to the mean field. The first is a decomposition: by the translation invariance of CVaR, writing the carbon field as mean plus zero-mean residual $\rho = \bar{\rho} + \xi$ and noting $\langle \bar{\rho}, x\rangle$ is deterministic,

$$\mathrm{CVaR}_\beta\big(\langle \rho, x\rangle\big) = \underbrace{\langle \bar{\rho}, x\rangle}_{\text{dependence-independent}} + \mathrm{CVaR}_\beta\big(\langle \xi, x\rangle\big), \tag{10}$$

so for a *fixed* schedule the dependence model moves only the residual term. The second is an a-priori bound on how far it can move the *optimal* value.

Proposition 1 (Spatial gap control, SOCP)**.** *Let* $\mathrm{OPT}(\Sigma)$ *be the optimal value of the SOCP* (2) *for a covariance* Σ*, and let* Σ^{shuf} *be its block-diagonal (cross-region-zeroed) counterpart with* $\Sigma^{\mathrm{off}} = \Sigma - \Sigma^{\mathrm{shuf}}$*. Then the spatial gap obeys*

$$\left|\mathrm{OPT}(\Sigma) - \mathrm{OPT}(\Sigma^{\mathrm{shuf}})\right| \;\leq\; \varepsilon \left(\max_{x\in\mathcal{X}} \|x\|_2\right) \left\|\Sigma^{\mathrm{off}}\right\|_2^{1/2}, \tag{11}$$

which vanishes as $\varepsilon \to 0$ *(no robustification) or* $\Sigma^{\mathrm{off}} \to 0$ *(no cross-region structure).*

Proof. The two programs share the feasible set $\mathcal{X}$ and differ only in the conic penalty. For fixed x, using $|\sqrt{a}-\sqrt{b}| \le \sqrt{|a-b|}$ and the Rayleigh bound $|x^\top \Sigma^{\mathrm{off}} x| \le \|\Sigma^{\mathrm{off}}\|_2 \|x\|_2^2$,

$$\left|\varepsilon\sqrt{x^\top \Sigma x} - \varepsilon\sqrt{x^\top \Sigma^{\mathrm{shuf}} x}\right| \le \varepsilon\sqrt{|x^\top \Sigma^{\mathrm{off}} x|} \le \varepsilon \|x\|_2 \|\Sigma^{\mathrm{off}}\|_2^{1/2}.$$

Since the infimum is 1-Lipschitz in the sup-norm of the objective, two objectives that differ pointwise by at most δ over a common feasible set have optimal values differing by at most δ. The feasible set $\mathcal{X}$ is a bounded polytope (the capacity bounds C0), hence compact, so the maximum $\max_{x\in\mathcal{X}}\|x\|_2$ is attained. Its value follows from combining the capacity bound $x_{r,t} \le \bar{x}$ (C0) with work conservation (C2b): $\|x\|_2^2 = \sum_{r,t} x_{r,t}^2 \le \bar{x}\sum_{r,t} x_{r,t} = \bar{x}\sum_r W_r = \bar{x}^2\, R\,T\,u$, so $\max_{x\in\mathcal{X}}\|x\|_2 \le \bar{x}\sqrt{R\,T\,u}$; taking that maximum gives (11). □

The bound is a conservative *certificate*, not a tight prediction. Evaluated on each grid's training covariance (with $\varepsilon^\star = 1$, the a-priori feasible bound $\max_x \|x\|_2 = \bar{x}\sqrt{R\,T\,u}$ where $\bar{x}$ is the per-cell cap and u the utilization, and $\|\Sigma^{\mathrm{off}}\|_2$ the spectral norm of the destroyed cross-region blocks), its right-hand side is 7.4% (Eastern US–Canada), 9.6% (Diversified) and 12.7% (Western US) of $\mathrm{CVaR}_{0.95}$; the realized gaps (Table 1) never exceed 0.23%, more than an order of magnitude smaller (a factor of 30–55 across the three grids). The theorem therefore bounds the spatial gap by a loose, non-binding worst-case certificate (right-hand side 7.4–12.7% of CVaR, 30–55× the realized gaps) *before any experiment is run*, and the experiments then pin it far below that cap. What the bound buys is not tightness but structure: the spatial gap is governed by ε and the cross-region block norm, and is forced to zero in the no-robustification and no-cross-structure limits. Most of the remaining slack lives in the feasible-diameter factor $\max_x \|x\|_2$, an a-priori worst case; substituting the realized optimal $\|x^\star\|_2$ would tighten the certificate considerably, at the cost of making it post-hoc rather than a-priori. We keep the a-priori form deliberately, since its purpose is to bound the gap *before* the experiment, not to predict it. The *tight* ceiling is measured, not bounded: the comonotone arm of Section 4.8 realizes the supremum over copulas, and across scenario seeds its gain is centred on zero with $|\Lambda| \lesssim 0.1\%$ of $\mathrm{CVaR}_{0.95}$, at the sampling-noise floor. The mean-leveling test of Section 4.6 bounds the same quantity from a second, independent angle. Both put Λ an order of magnitude below the schedule's mean-driven value; hence the mean field pins the schedule and the dependence model, elliptical or not, is immaterial. The prediction that even Clayton cannot recover material value is confirmed in Section 4.8.

Computational considerations. Each instance is small: the decision vector has $RT \le 5 \times 24 = 120$ entries and the feasible set adds $O(RT)$ linear constraints, so the SOCP (2) solves in 0.19–0.26 s and the CVaR-SAA LP (9), which appends $S = 1000$ epigraph variables and constraints, in 0.9–1.3 s (median over five repetitions, CLARABEL/HiGHS, commodity laptop). The cost is in the *number* of instances, not their size: one region-set experiment solves 3 regimes × 3 α × 2 arms ×

($6\,\varepsilon \times 5$ folds + 1 test read) = 558 SOCPs, and the full Phase 1 battery (region sets $\times$ {sample, Ledoit–Wolf, seasonal, AR(1), two mean-leveling settings, walk-forward, fine grid}) is on the order of 10^4 SOCP solves; Phase 2 adds $3 \times 3 \times 3 \times 4 = 108$ CVaR-SAA LPs. Total wall-clock for the entire study is under two hours on one core, so reproduction is cheap.

3.11 Reproducibility

All code is version-controlled (private repository, continuous-integration test suite, 199 unit tests, including a feasible-set equivalence test pinning the Phase 2 CVaR scheduler to the Phase 1 constraint set); experiment configurations are commit-locked before test-set evaluation; bootstrap and scenario seeds are fixed; and every number reported in Sections 4–5 traces to an archived summary table. Solvers: CLARABEL for the SOCP; HiGHS for the CVaR-SAA linear programs and deterministic LP baselines.

4 Results

4.1 The spatial assumption is valid (RQ1)

The premise holds. After removing each zone's hour-of-day mean, cross-region correlations of carbon intensity in the two electrically coupled US/Canada cases are strong and survive de-seasonalization: in the US West, CISO–NEVP 0.78 and CISO–LDWP 0.72 (shared solar resource and weather across the Western Interconnection); in the Ontario–Eastern belt, CA-ON–NYISO 0.73 and MISO–PJM 0.70 (weather fronts traversing the Eastern Interconnection). The engineered portfolio behaves as designed, but the design is *heterogeneity*, not zero correlation: its three residual pairs span −0.01 (ERCOT–BPAT, wind vs. hydro), 0.29 (CISO–ERCOT) and 0.37 (CISO–BPAT), mean $\bar{r} \approx 0.2$. It is therefore the low, *mixed-correlation* end of the US cases, one near-zero edge and two moderate ones, a diversified generation mix rather than a uniformly decoupled grid. This distinction matters later: heterogeneous, partly low correlation is *diversifiable* structure (it drives the largest mean-leveling test signal, Section 4.6), whereas the US West's strong but *uniform* correlation is common-mode and offers no hedge. Figure 3 shows the residual correlation matrices; overlay plots of standardized intensity (archived with the code) confirm visible co-movement in summer and winter weeks for the coupled cases. Together with the corroborating panels, the California/Nevada subset (0.65–0.78 within-state) and Iberia–France, the cases span the dependence spectrum from ≈ 0 to ≈ 0.8, so whatever the scheduling result, it cannot be attributed to an absence of correlation in the data.

Figure 4 places the three grids geographically and draws each within-grid residual correlation as

Figure 3: Residual (hour-of-day-removed) cross-region correlation of hourly carbon intensity, 2021–2025. Left: US West (CISO, BANC, LDWP, NEVP, AZPS). Right: Ontario–Eastern belt (CA-ON, NYISO, MISO, PJM). Cross-region correlation is strong and genuine in both, the assumption motivating spatial DRO is empirically valid.

a weighted edge. The picture matches the physics: the US West forms one tightly-coupled Western Interconnection cluster, the Ontario–Eastern belt another along the Eastern Interconnection, while the engineered portfolio spans the continent with deliberately *mixed* edges, one near-zero and two moderate, by construction.

4.2 The day-ahead scheduler and its savings (RQ1)

The scheduler migrates compute across regions against a day-ahead forecast. We report out-of-sample $\mathrm{CVaR}_{0.95}$ on the *same* feasible set against the honest comparator: carbon-aware scheduling with no inter-region transfer ($\Phi = 0$), each region scheduled by its own forecast with no inter-region flow. The transfer arm adds flows $f_{r\to s,t} \geq 0$ under a budget Φ.

Figure 5 reports the saving against Φ. Active transfer cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% relative to the $\Phi = 0$ baseline across the three grids (Western 4.0%, Eastern 9.9%, Diversified 9.0%). The lever is the spatial *mean*: at each hour one region is cleanest on average, and migration ships work there. The saving is *deterministic*, captured without any dependence model. This is consistent with the carbon-computing literature, which finds spatial shifting dominates temporal shifting [20, 23]. We deliberately report the saving over the $\Phi = 0$ carbon-aware baseline (not over a uniform spread or a carbon-blind schedule): the honest lever is transfer versus no-transfer. The out-of-sample reduction is largest at a moderate budget (on the US West a 4.4% peak near 10% of daily workload) and settles to the reported 4.04% plateau as the budget grows: at large budgets the deterministic schedule reallocates further toward the training-mean-cleanest region and slightly overfits out of sample, so we quote the conservative plateau as the headline rather than the peak. Section 6 develops the full transfer DRO, its multi-seed validation, and a data-grounded emergency stress test.

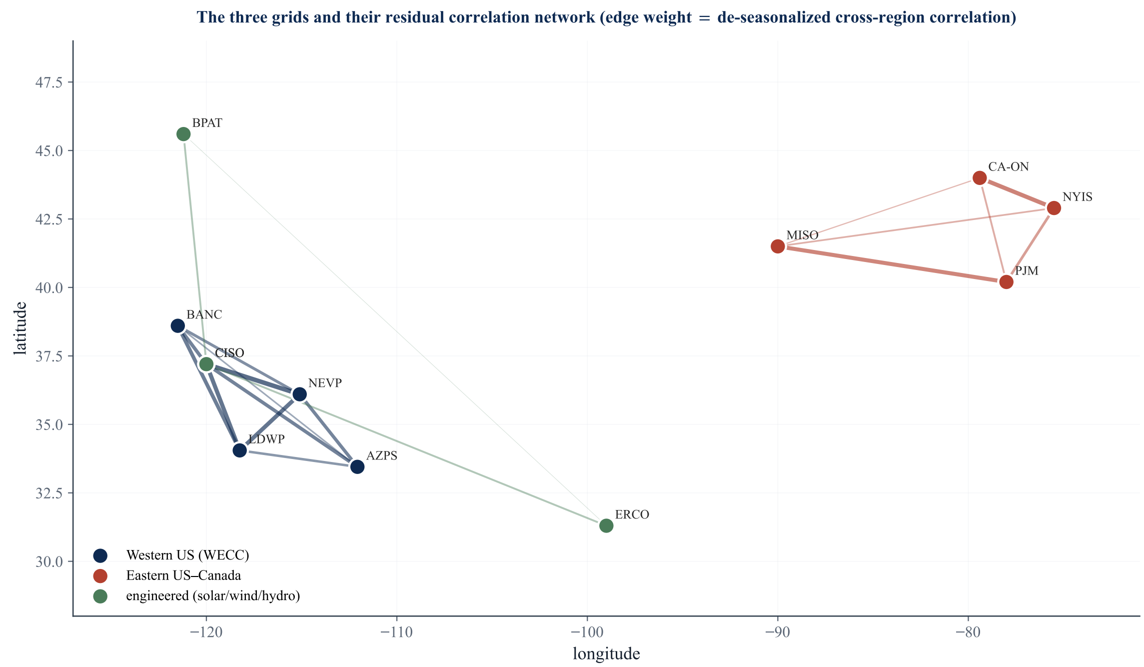

Figure 4: The three grids and their residual correlation network. Nodes are Electricity Maps zones at their geographic centroids; edge weight is the de-seasonalized cross-region correlation. Tight clusters (US West, Ontario–Eastern belt) contrast with the deliberately diversified engineered portfolio.

4.3 Where the value comes from: the complexity–value frontier (RQ1)

Figure 6 places every model class on a single complexity–value frontier (carbon-blind, deterministic per-region, robust per-region, deterministic transfer, robust transfer). Deterministic transfer dominates the frontier; the passive covariance and copula layers add no height (Sections 4.4–4.8); robust transfer pays only in the tail (Section 4.9). Three forces explain why the transfer saving is mean-driven and why passive covariance cannot add to it: the within-day swing of mean intensity dwarfs the residual covariance; the strongest correlation (US West) is common-mode and so offers no hedge; and the empirical upper tail de-couples (χ_{U} at or below Gaussian) while the co-movement that exists is in the *clean* tail ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), which a risk-averse scheduler is not protecting against and an elliptical ball cannot encode. The mechanism is developed in Section 5.2.

4.4 The screening rule: passive covariance adds nothing (RQ2)

Table 1 reports the spatial gap (8) for the three pre-registered US/Canada cases: nine cells each (three constraint regimes × three α-levels), CV-selected $\varepsilon^{\star}$, single test read on 2025. The gap never exceeds a few tenths of one percent of test $\mathrm{CVaR}_{0.95}$ in either direction. Where the bootstrap calls a cell "detectable" the effect is genuine but economically negligible (on the order of a few hundred

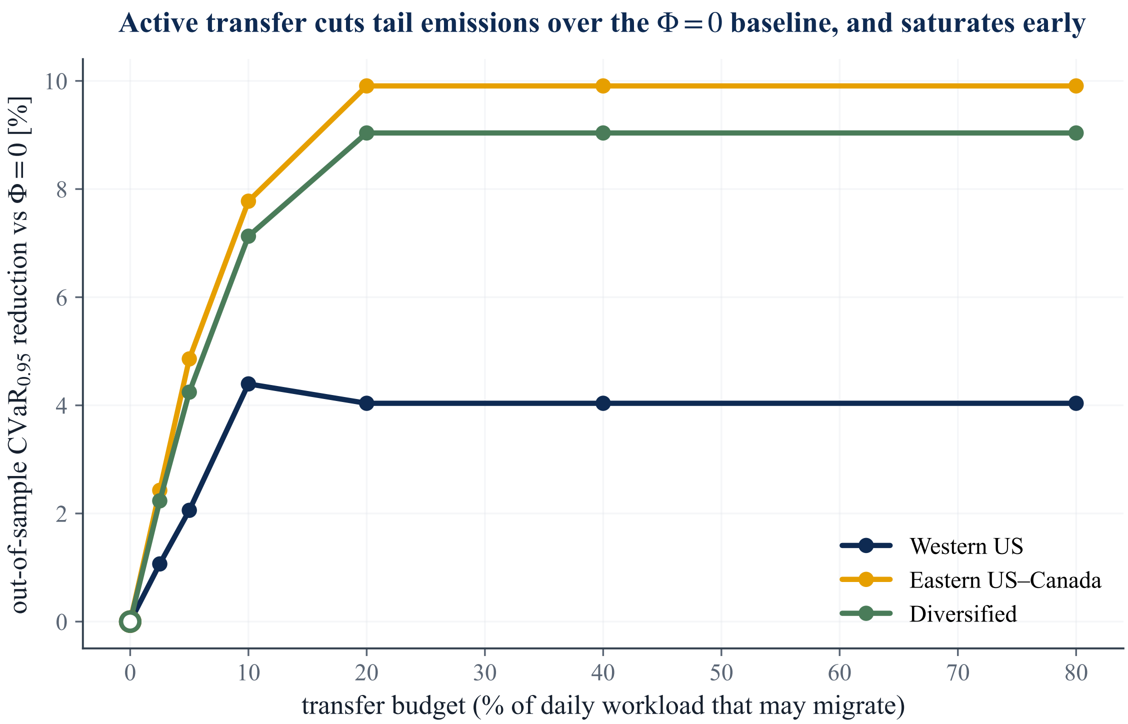

Figure 5: The transfer lever. Out-of-sample $\mathrm{CVaR}_{0.95}$ saving against the inter-region transfer budget Φ, relative to the carbon-aware no-transfer baseline ($\Phi = 0$). Active migration captures this spatial lever; the curve saturates as the feasible mean-based gain is exhausted. The robust layer is analyzed in Section 4.9.

$\mathrm{gCO_2eq/kWh}$ out of a CVaR of 1.5×10^6) and, decisively, its *sign* flips across adjacent α-levels within the same regime. A consistently exploitable spatial structure would not reverse direction under a change of the flexible-load fraction; these reversals are the signature of optimization at the boundary of materiality, not of a usable effect. (We confirmed the detectable cells are not numerical: re-solving with CLARABEL and SCS reproduces the gap to four significant figures, so the sign instability is in the data, not the solver.) The California/Nevada panel agrees (gaps in $[-0.012, +0.058]\%$, DRO active throughout).

Crucially, this is a null of the *strong* kind. In all three pre-registered cases cross-validation *chooses* a non-trivial ambiguity radius: $\varepsilon^\star = 1$ in all nine cells for both Eastern US–Canada and Diversified, and in seven of nine for Western US (25 of the 27 cells overall). The DRO is therefore genuinely engaged (it is paying for robustness, not collapsing to the nominal problem) and the two arms still coincide. A null in which CV had selected $\varepsilon^\star = 0$ would be uninformative (robustification switched off, so the arms must match by construction); here the robustification is demonstrably on and the cross-region covariance still buys nothing. This is the first regime in the project where the DRO is consistently active, which is precisely what makes the null informative rather than vacuous. Figure 7 shows the cross-validation curves directly: validation $\mathrm{CVaR}_{0.95}$ dips

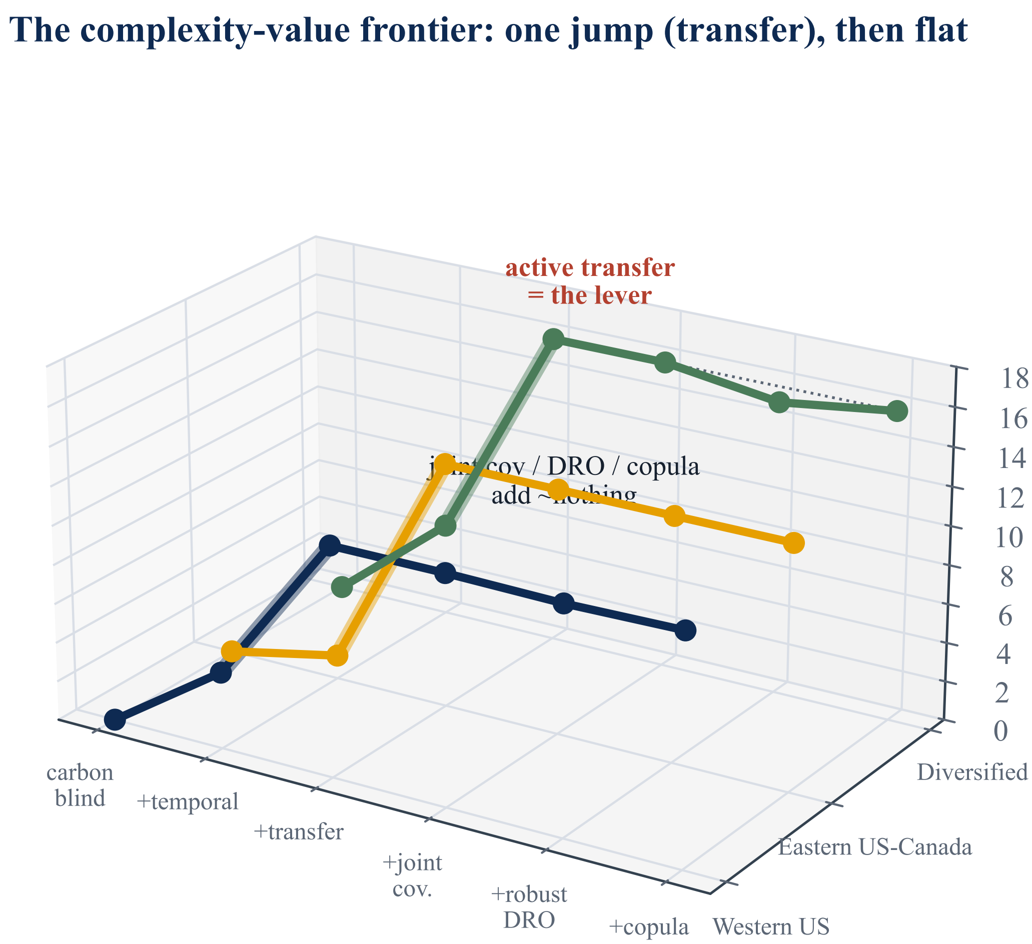

Figure 6: The complexity–value frontier. Out-of-sample $\mathrm{CVaR}_{0.95}$ saving against modelling and estimation complexity. The value lives in deterministic transfer; passive dependence sophistication is flat; robust transfer is conditional on tail severity (Section 4.9).

to an interior minimum at a non-trivial $\varepsilon^\star$ before rising as over-conservatism sets in, and the joint and shuffled curves lie on top of one another.

The Iberia–France panel is the instructive contrast and the project's historical starting point. There the residual correlation is near zero, and CV does not even consistently engage the DRO: $\varepsilon^\star$ flickers between 0, 0.1, 1 and 10 across cells, collapsing to 0 in several. This is itself consistent with the thesis (where there is no cross-region structure, the machinery sees nothing worth robustifying) but it makes the Iberian null the *weak* kind. We therefore retain it only as low-correlation, real-grid external validity, and rest the central claim on the US/Canada cases where the DRO is unambiguously on. The engineered Diversified portfolio supplies the DRO-active low-correlation null ($\varepsilon^\star = 1$ throughout, still no value) that Iberia, where $\varepsilon^\star$ collapses to 0, cannot.

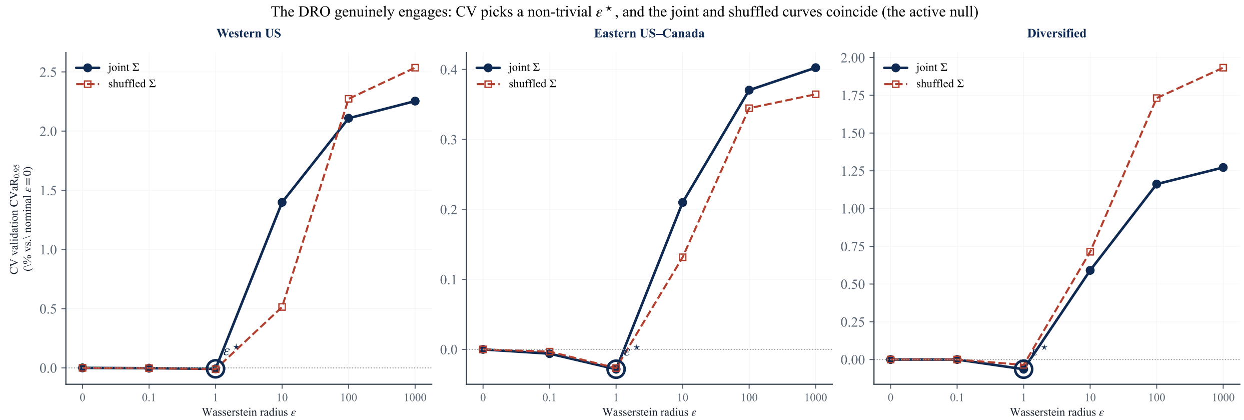

Figure 7: The DRO genuinely engages. Blocked 5-fold cross-validation $CVaR_{0.95}$ (relative to the nominal $\varepsilon = 0$ problem) against the Wasserstein radius ε, for the joint (solid) and shuffled (dashed) covariances. Cross-validation selects an interior $\varepsilon^\star$ (circled), not $\varepsilon = 0$, so the robustification is on; the two curves coincide, so the cross-region covariance changes nothing. Large ε is over-conservative.

Three readings of Table 1 matter. First, in the adversarial diversification case (Diversified) the joint covariance actually *hurts* at low α (up to -0.23%): with no true cross-structure, the joint estimator fits noise that the block-diagonal arm is spared. Second, in the strongly correlated belt (Eastern US–Canada) the detectable cells flip sign between $\alpha = 0.30$ and $\alpha = 0.75$ within the same regime. Third, the US West (strongest correlation of all) produces no detectable cell beyond one of magnitude $+3 \times 10^{-6}\%$. Figure 8A plots all 27 cells against each case's mean residual correlation: the cloud stays on zero across the whole correlation range. **Validity of the spatial assumption does not imply value from it.**

The null is visible in the schedules themselves. Figure 9 overlays the optimal load for the joint and shuffled covariances on each region's mean carbon field: the two schedules are indistinguishable to the eye, and both place compute in the cleaner hours subject to the deadline and ramp constraints. Destroying the cross-region covariance changes essentially nothing about what the scheduler does.

4.5 The null survives a pre-registered robustness battery (RQ2)

Table 4 summarizes. Re-estimating the covariance with Ledoit–Wolf shrinkage, seasonal residuals, or AR(1) residuals never produces a spatial effect that passes the pre-registered agreement rule (positive and sign-stable across both residual estimators). The largest battery value anywhere, $+0.179\%$ (AR(1), Eastern US–Canada, $\alpha = 0.30$), appears *only* under AR(1) and not under the seasonal estimator, so the rule rejects it. Benjamini–Hochberg correction at $q = 0.05$ across all 144 real-data cells (four grid-runs, the three primary grids plus the locked Task C replication, each at 3 regimes $\times$ 3 α = 9 cells under the four covariance estimators {sample, Ledoit–Wolf, seasonal,

Table 1: Spatial gap (% of test $\mathrm{CVaR}_{0.95}$; positive = joint covariance better) for the three US/Canada cases. R3 = reference regime, R1 = +deadline, R2 = +variable carbon-coupled capacity. "det." = bootstrap 95% CI excludes zero.

		Diversified ($\bar{r} \approx 0.2$)		Eastern US–Canada ($\bar{r} \approx 0.45$)		Western US ($\bar{r} \approx 0.65$)	
Regime	α	gap %	det.	gap %	det.	gap %	det.
R3	0.30	−0.233	✓	−0.022	✓	+0.043	
R3	0.50	−0.177	✓	−0.002		+0.032	
R3	0.75	+0.001		+0.034	✓	−0.005	
R1	0.30	−0.233	✓	−0.022	✓	+0.014	
R1	0.50	−0.177	✓	−0.002		+0.028	
R1	0.75	+0.001		+0.034	✓	+0.002	
R2	0.30	−0.211	✓	+0.038	✓	+0.008	
R2	0.50	−0.154	✓	+0.030	✓	+0.005	
R2	0.75	+0.062	✓	+0.009		+0.000	✓

AR(1)}: $4\times9\times4 = 144$) leaves that same already-rejected cell as the largest surviving positive gap; nothing material survives both filters. A walk-forward refit (train 2021–2023, test 2024) reproduces the null out of time: maximum absolute gap 0.036% (Eastern US–Canada), 0.028% (Western US), 0.217% (Diversified, again negative). An 11-point log-denser ε grid changes no gap at the reported precision. Finally, a *constraint-tightness* sensitivity directly probes the most plausible route to spatial value: tripling the binding tightness of the ramp limit ($\Delta_r = 5$ instead of 15 MW/h) shrinks the feasible set so the schedule has far less freedom to track the mean field, yet the null is unmoved, with maximum absolute gap 0.028% (Western US), 0.044% (Eastern US–Canada) and 0.169% (Diversified), all sign-flipping and with $\varepsilon^\star = 1$ retained. The covariance does not begin to pay even when the mean's pull is mechanically curtailed. A complementary *utilization* sweep makes the same point through the workload: raising daily utilization from the baseline 80% to 95% packs the facility near capacity and strips out almost all scheduling freedom, yet the gap only *shrinks* (maximum absolute gap 0.017% for Western US, 0.048% for Eastern US–Canada, 0.071% for Diversified), and at a loose 50% it stays immaterial ($\leq 0.25\%$); $\varepsilon^\star = 1$ throughout. Across the full $\{50, 80, 95\}\%$ range the null is unmoved. Figure 10 collects the covariance-estimator and temporal battery: across all six arms and three grids, every spatial gap sits inside the no-value band.

Tail level: the null holds from $\mathrm{CVaR}_{0.90}$ to $\mathrm{CVaR}_{0.99}$. The primary read fixes the tail at $\beta = 0.95$. Because the residual co-movement lives in the *clean* tail (Section 4.7), the natural worry is that spatial value might surface only at a deeper risk level. Re-running the full pipeline (cross-validated ε, single 2025 read) at $\beta \in \{0.90, 0.95, 0.99\}$ rejects this (Table 2). The DRO stays engaged ($\varepsilon^\star = 1$ almost everywhere) and the gap stays immaterial at every level. Probing the deepest tail widens the magnitudes slightly but never to materiality: the largest *positive* gap anywhere is +0.14%

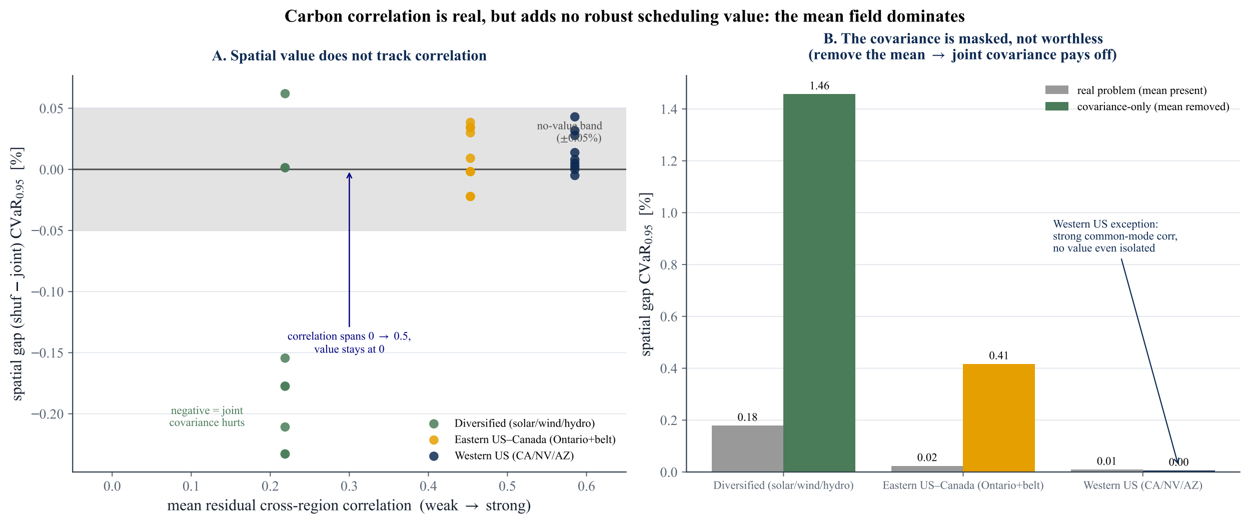

Figure 8: The Phase 1 finding. **A:** spatial gap vs. mean residual cross-region correlation; value does not track correlation anywhere on the spectrum. **B:** mean-leveling test; in the covariance-only world (constant scheduling mean) the joint covariance pays off by up to +1.46%, proving the spatial signal is real but *masked* by the mean carbon field, not absent.

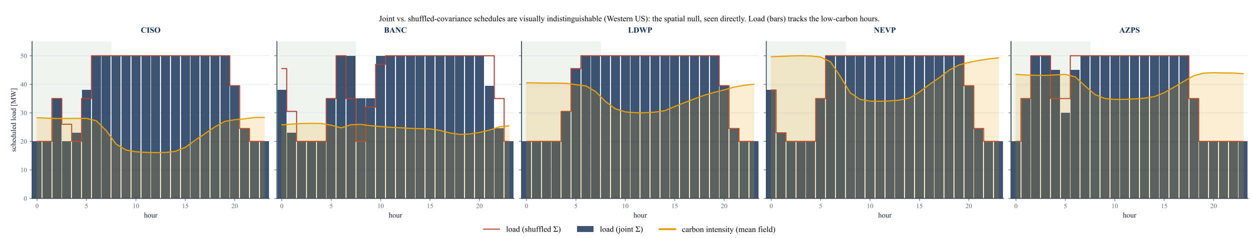

Figure 9: The spatial null, seen directly (Western US, regime R3, $\alpha = 0.5$, $\varepsilon^\star = 1$). Bars are the optimal load under the joint covariance; the red step is the load under the shuffled (block-diagonal) covariance. The two coincide. The gold curve is each region's mean carbon intensity; the green band marks the deadline window $[0, 7]$.

(Western US, $\beta = 0.99$), still a third of the 0.4% margin and not sign-stable, while in the adversarial Diversified grid the joint covariance becomes *more* negative at $\beta = 0.99$ (−0.36%, i.e. it hurts), exactly as the noise-fitting account predicts. Deepening the tail, the regime where elliptical-blind tail dependence should matter most, does not turn the covariance on. The $\beta = 0.99$ estimates are necessarily noisier, since the tail thins to a handful of days: the per-cell paired-bootstrap standard error rises to 0.15% for Western US (against ≤ 0.04% at $\beta = 0.90$), so that grid's largest $\beta = 0.99$ point estimate (+0.14%) is itself within one standard error of zero. The apparent widening at the deepest tail is sampling noise, not emerging spatial value. This sweep is reported as a descriptive robustness check, *outside* the pre-registered 144-cell FDR family of Table 4; for completeness, no sweep cell produces a gap that is both positive and materially large (every detectable cell is either negative or below the 0.4% margin), so none would survive multiple-testing correction as a

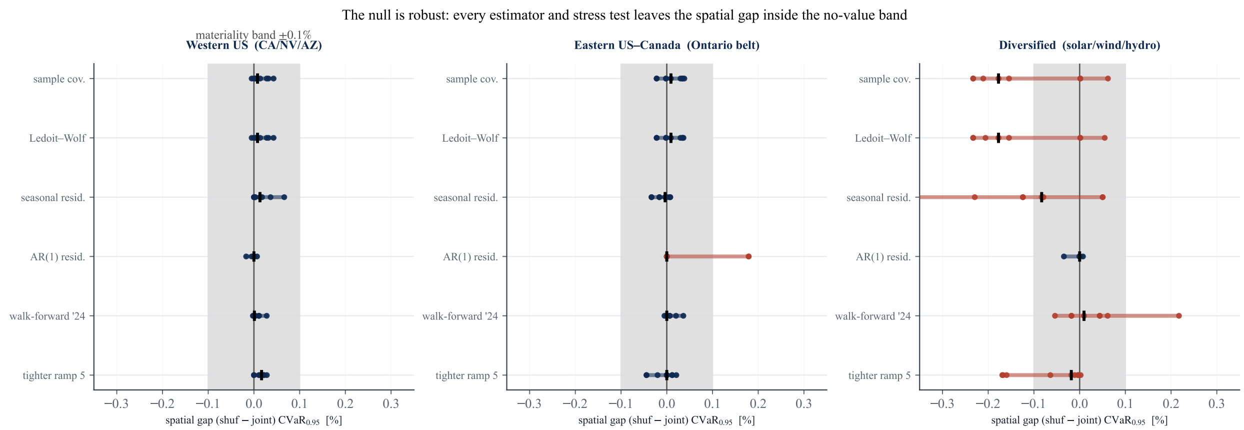

Figure 10: Robustness of the null. For each grid and each robustness arm (sample covariance, Ledoit–Wolf, seasonal and AR(1) residuals, walk-forward to 2024, and a 3× tighter ramp), dots are the spatial gap for the nine (regime, α) cells and the tick marks the median. The shaded band is the ±0.1% no-value threshold. No arm produces a sign-stable, material gap; the few Diversified excursions are negative (the joint covariance mildly hurts).

discovery.

Table 2: Spatial gap range (min, max over the nine cells, % of test CVaR$_\beta$) at three tail levels. The DRO is engaged ($\varepsilon^\star = 1$ almost everywhere) throughout. No level yields a sign-stable positive gap reaching the 0.4% materiality margin.

Grid	$\beta = 0.90$	$\beta = 0.95$	$\beta = 0.99$
Western US	[−0.04, +0.01]	[−0.01, +0.04]	[0.00, +0.14]
Eastern US–Canada	[−0.02, +0.04]	[−0.02, +0.04]	[−0.01, +0.09]
Diversified	[−0.23, +0.08]	[−0.23, +0.06]	[−0.36, +0.02]

The null is not a conditioning artifact. Block-diagonalizing Σ also changes its conditioning, so one might worry the null reflects the joint arm being harder to estimate rather than the cross-structure being valueless. The two arms are conditioned to the same order (ridge-regularized $\kappa \approx 1.1$–1.6×10^5 for the joint covariance, 0.4–0.8×10^5 for the shuffled, the joint arm somewhat higher as expected for the richer model). The mean-leveling test of Section 4.6 is the decisive control: it reuses the *same* joint Σ at the *same* conditioning and, removing only the mean field, the covariance recovers up to +1.46%. Estimation conditioning therefore cannot be what suppresses the spatial value; the mean field is.

Statistical power: the null is not an underpowered test. A null is only as strong as the test's ability to reject it, so we quantify that ability directly. The paired day-bootstrap gives a standard error on the spatial gap of SE = 0.005–0.039% of $\mathrm{CVaR}_{0.95}$ across the three primary grids (Table 3). The minimum spatial benefit detectable at 80% power (two-sided, $\alpha = 0.05$) is $\mathrm{MDE} = (z_{0.975} + z_{0.80})\,\mathrm{SE} = 2.80\,\mathrm{SE} = 0.015$–0.11%. The test therefore had the power to detect a spatial advantage *an order of magnitude below* a $\approx 1\%$ materiality threshold (the smallest gap a practitioner would act on; by the economic grounding of Section 5.4 this is $\approx 0.4\%$ of $\mathrm{CVaR}_{0.95}$), and roughly the same order as the per-cell gaps themselves, yet no sign-stable positive gap of even this size survives. The null is a genuine absence of usable spatial value, not a test too blunt to see one.

Equivalence: the absence of a material effect is itself significant. Power bounds the effect the test *could* have missed; an equivalence test makes the complementary, affirmative claim that a material effect is *absent*. We apply the two-one-sided-tests (TOST) procedure [25, 26] *per cell*, using each cell's own paired-bootstrap standard error (recovered from its 95% gap interval), against the economic materiality margin $\Delta = 0.4\%$ of $\mathrm{CVaR}_{0.95}$ (Section 5.4; the $\approx 1\%$ practitioner threshold is looser still), testing $H_0\colon |\mathrm{gap}| \geq \Delta$ against $H_1\colon |\mathrm{gap}| < \Delta$. All 27 cells across the three grids are equivalent: every 90% confidence interval lies strictly inside $(-\Delta, +\Delta)$. The binding cell is the adversarial Diversified case (gap = −0.233%, cell SE = 0.046%), which still clears both one-sided tests, $z = (\Delta - |\mathrm{gap}|)/\mathrm{SE} = 3.7$ ($p = 1.3 \times 10^{-4}$); every other cell gives $z \geq 7$. We therefore do not merely fail to detect a spatial benefit; we statistically reject the existence of one (in either direction) larger than the materiality threshold, cell by cell. That is the strongest form a null can take.

The claim is of course threshold-sensitive, so we state where it breaks. The smallest margin each grid still clears (its largest $|\mathrm{gap}| + 1.645\,\mathrm{SE}$ over cells) is $\Delta^\star = 0.05\%$ for Eastern US–Canada and 0.10% for Western US, but $\Delta^\star = 0.31\%$ for the adversarial Diversified grid. Equivalence at the 0.4% economic margin therefore holds everywhere with room to spare on the two interconnected grids, and only narrowly on the engineered low, heterogeneous-correlation portfolio. Equivalently, the conclusion holds for any materiality margin $\Delta \geq 0.31\%$; the 0.4% figure is fixed a priori from the economic grounding of Section 5.4, not reverse-engineered to clear the test. That last caveat does not threaten the thesis: the Diversified gap is *negative*, so the one grid where equivalence is hardest to assert is precisely the one where the joint covariance *hurts* rather than adds value.

4.6 Why the covariance is masked: the mean-leveling test (RQ2)

The ablation isolates the cause (Table 5, Figure 8B). Under the LEVEL setting (regional time-averages equalized, hour-of-day shape kept) every gap is unchanged to three decimals: the *between-region* mean differences are not what masks the covariance. Under the FLAT setting (constant scheduling

Table 3: Statistical power of the spatial-gap test. SE is the paired-bootstrap standard error of the gap; $\mathrm{MDE}_{80} = 2.80\,\mathrm{SE}$ is the smallest true gap detectable at 80% power ($\alpha = 0.05$, two-sided). All are well below the $\approx 1\%$ materiality threshold.

Grid	SE (% $\mathrm{CVaR}_{0.95}$)	MDE_{80} (%)
Western US	0.036	0.10
Eastern US–Canada	0.005	0.015
Diversified	0.039	0.11

Table 4: Robustness battery: maximum |gap| (%) per covariance estimator, and temporal walk-forward (test year 2024). No cell passes the pre-registered agreement rule (positive and sign-stable under both seasonal and AR(1) residualization).

Case	Ledoit–Wolf	Seasonal	AR(1)	Walk-forward 2024
Diversified	0.233	0.355	0.035	0.217
Eastern US–Canada	0.037	0.034	0.179	0.036
Western US	0.043	0.066	0.017	0.028

mean, the covariance-only world) the joint covariance suddenly pays: up to +1.46% in Diversified and +0.41% in Eastern US–Canada, all BH-significant, and CV no longer collapses both arms onto the same schedule. The spatial signal in the covariance is therefore *real and exploitable*; in the real problem it is simply dominated by the within-day shape of the mean carbon field, which both arms share. The instructive exception is Western US: even in the covariance-only world its gaps stay at zero (−0.036 to +0.004%), because its strong correlation is *common-mode*, all five zones move together, so destroying cross-dependence changes no scheduling decision. Positive correlation is non-diversifiable risk: there is no hedge for a DRO to find.

The flat-setting result doubles as the test's *positive control*: in a world constructed so the dependence layer *should* pay, the same protocol fires (up to +1.46%), whereas on the real data it reads zero. That contrast is what licenses the main null as a true negative, a genuine absence of value, rather than a test too blunt to detect it.

4.7 Tail dependence: the structure is non-elliptical (RQ2)

Table 6 reports residual tail-dependence at $q = 0.95$ for representative pairs. Two findings. *First*, the upper tail is empirically at or below the Gaussian benchmark everywhere (excess −0.18 to +0.04): in the dirtiest hours, regions *de-couple* rather than spike together, there is no upper-tail co-movement for a richer ambiguity set to exploit on the risk side. *Second*, the dependence is radially asymmetric, $\chi_L > \chi_U$: regions go *clean* together more strongly than dirty together, clearest in the solar-driven US West (CISO–LDWP $\chi_L = 0.48$ vs. $\chi_U = 0.25$; BANC–LDWP 0.40 vs.

Table 5: Mean-leveling test: spatial gap range (%) across the nine cells per case. LEVEL (equalize regional time-averages) leaves the null untouched; FLAT (constant scheduling mean = covariance-only world) reveals genuine spatial value except where correlation is common-mode (Western US).

Case	Real problem	LEVEL	FLAT (covariance-only)
Diversified	$-0.233\ldots+0.062$	unchanged	$+0.39\ldots+1.46$
Eastern US–Canada	$-0.022\ldots+0.038$	unchanged	$+0.14\ldots+0.41$
Western US	$-0.005\ldots+0.043$	unchanged	$-0.04\ldots+0.00$

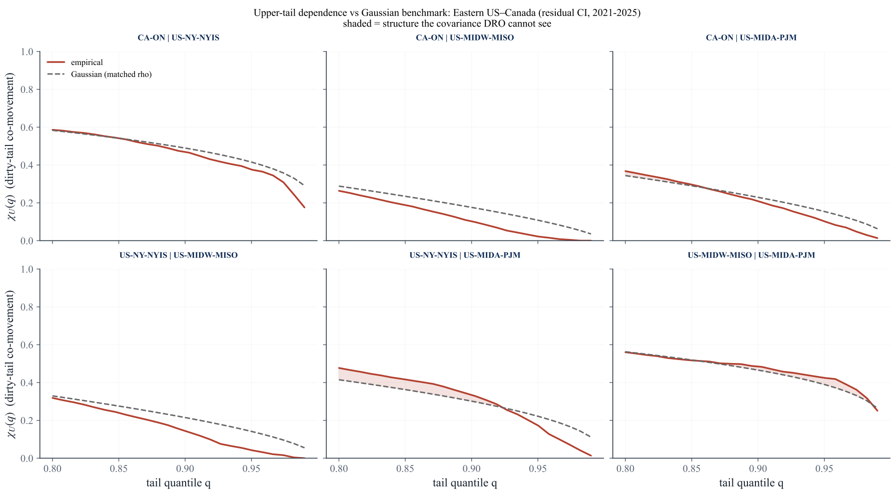

Figure 11: Upper-tail dependence $\chi_U(q)$ for the Ontario–Eastern belt region pairs (solid) against the Gaussian-copula benchmark at the matched correlation (dashed). The empirical curves sit at or below the benchmark as $q \to 1$: in the dirtiest hours the regions decouple, so there is no exploitable upper-tail co-movement for a richer ambiguity set to capture. The complementary radial asymmetry ($\chi_L > \chi_U$) is quantified in Table 6.

0.17). Elliptical copulas force $\chi_L = \chi_U$, so this is structure a Mahalanobis–Wasserstein ball is geometrically incapable of representing, the covariance is not merely masked (Section 4.6); for the part of the dependence that varies, it is also the *wrong object*. Figure 11 shows the asymmetry for the Ontario–Eastern belt.

4.8 Richer dependence models confirm the screening rule (RQ2)

If the covariance ball failed only because it is elliptical, a copula that encodes the empirical $\chi_L > \chi_U$ asymmetry should recover value. It does not. Re-running the falsification with Gaussian, Clayton, and the maximal (comonotone) copula on the same feasible set leaves the screening rule intact: across scenario seeds even the maximal coupling's gain is centred on zero at the sampling-noise floor ($|\Lambda| \lesssim 0.1\%$ of $\mathrm{CVaR}_{0.95}$). The binding constraint is the mean field, not the shape of the dependence object, exactly as Proposition 1 predicts. Full protocol, tables, and figures are in

Table 6: Residual tail dependence at $q = 0.95$ for selected pairs. "Excess" = empirical χ_U minus the Gaussian-copula value at the matched correlation ρ. Throughout, χ_U shows no exploitable upper-tail excess while $\chi_L > \chi_U$ signals non-elliptical (radially asymmetric) dependence.

Case	Pair	ρ	χ_U emp	χ_U Gauss	Excess	χ_L emp
Western US	CISO–LDWP	0.72	0.25	0.41	−0.16	0.48
Western US	BANC–LDWP	0.65	0.17	0.35	−0.18	0.40
Western US	CISO–NEVP	0.78	0.43	0.47	−0.05	0.39
Eastern US–Canada	CA-ON–NYISO	0.73	0.38	0.42	−0.04	0.31
Eastern US–Canada	MISO–PJM	0.70	0.43	0.39	+0.04	0.42
Diversified	CISO–BPAT	0.33	0.10	0.16	−0.06	0.26
Diversified	ERCOT–BPAT	0.00	0.04	0.05	−0.02	0.02

Appendix B.

4.9 The price of robustness: when does the crossover pay? (RQ3)

Passive sophistication adds nothing under nominal conditions; the honest question the title poses is when *robustness* pays. We make the spatial structure an active decision (inter-region flows) and add a two-stage robust commitment with migration-limited recourse, hedging day-ahead forecast error. By Proposition 1's logic the mean dominates under nominal carbon, and indeed robust ≈ deterministic there: the DRO buys no mean reduction (a price-of-robustness result, not an emissions win). The value of the stochastic solution (VSS, how much planning against the full uncertainty distribution beats committing to a single forecast) is near zero at best. In the rolling online evaluation (Table 8) the robust controller's out-of-sample $\mathrm{CVaR}_{0.95}$ is statistically indistinguishable from the deterministic one on the US West (+0.28%, 95% CI [−0.65, +1.16]) and Eastern (+0.04%, [−0.10, +0.25]) grids, but is significantly *worse* on the Diversified grid (−10.34%, [−18.3, −4.4], a single representative seed; the four-seed mean is a milder but still negative −4.4%, Section 7): hedging forecast error when no emergency is present costs tail performance where the regions are weakly coupled. This sharpens the decision rule rather than softening it: under nominal conditions robustness is at best free and can actively hurt, so it is worth buying only when emergencies past the crossover are expected.

The null holds in the deep tail. A robust controller is a tail hedge, so a natural objection is that $\mathrm{CVaR}_{0.95}$ understates its value for a more tail-averse operator. We re-evaluated the same online controllers at progressively deeper risk levels, from $\mathrm{CVaR}_{0.90}$ through $\mathrm{CVaR}_{0.99}$ to the single worst day (the `dro_tail_sensitivity` snapshot). The gap stays immaterial at every level: none of the twelve grid-by-metric cells shows a statistically significant robust improvement, and on the Diversified grid robustness stays worse. The mean-dominance ceiling therefore holds beyond the headline tail level; day-ahead robustness does not earn its keep even for a maximally tail-averse operator.

Robustness earns its keep only in the tail. Modelling rare grid-emergency events (with probability 0.1 per day a region's carbon scales by a severity M), the robust commitment's advantage over the risk-neutral one grows with M. Across four independent emergency-scenario seeds the crossover survives on the strongly correlated US West grid only: the gain is -0.4% at $M = 1$, $+0.6\%$ (all four seeds) at $M = 2$, and $+4.4\%$ of $\mathrm{CVaR}_{0.95}$ on average (s.d. 3.4%, three of four seeds) at $M = 4$ (Figure 12). On the Eastern and Diversified grids the gain hovers near zero with unstable sign and does not survive multi-seed testing, so the single-seed positives there are scenario-seed noise (Section 6 gives the full four-seed treatment). The crossover is therefore real but grid-specific and only marginally significant (US West, one-sided), with $M^\star \approx 3$ the materially relevant threshold. The robust schedule stays hedged while the risk-neutral one is burned when its favoured region spikes.

Real grids stay below the crossover. The synthetic multiplier overstates real risk, and a data-grounded test confirms the crossover does not activate. On the *joint* (portfolio) axis the crossover lives on, the worst realized day reaches only 1.3–1.9× (joint portfolio axis) the portfolio mean across the three grids (Western 1.34×, Eastern 1.29×, Diversified 1.89×; archived in `carbon_ceiling.csv`), well below $M^\star \approx 3$. Per-region *relative* swings can be larger on very clean grids (BPAT 5.1×, Ontario 2.6×), but those sit on near-zero carbon bases and do not move the portfolio tail, so they are not joint emergencies; Winter Storm Uri, the worst US grid event on record, reached only 1.3×. Decisively, when emergencies are drawn *from real data* (each emergency replaces a region's day with a draw from its own empirical top-5% tail; `run_part3_real_emergency.py`), the robust commitment does *not* pay: its $\mathrm{CVaR}_{0.95}$ gain over the risk-neutral schedule is negative or within noise on every grid and frequency (Western -0.3%, Diversified -0.9% to -2.3%, Eastern $+0.1\%$ to $+0.2\%$). Real emergencies are not severe enough, in joint absolute-emission terms, to reach the crossover, so deterministic transfer is dominant on observed data. We claim no universal carbon-ceiling impossibility, only the tested one: across the zones and real emergencies we have, robustness does not activate. Section 6 gives the full multi-seed treatment and the data-grounded emergency model behind this verdict.

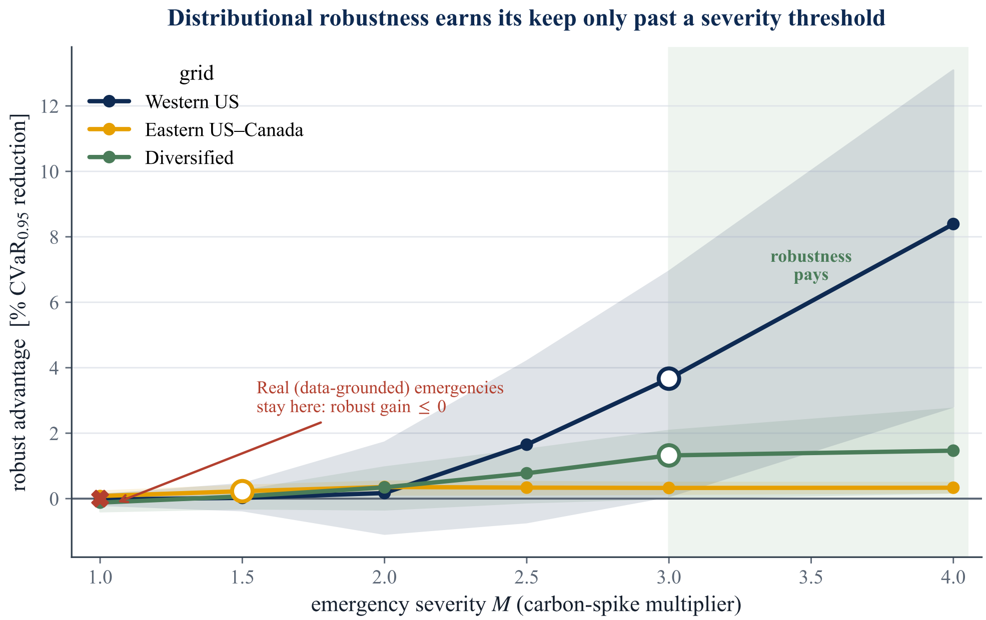

Figure 12: The price-of-robustness crossover. Robust-minus-risk-neutral $\mathrm{CVaR}_{0.95}$ advantage against emergency severity M. The materially relevant crossover is $M^\star \approx 3$ (grid-specific); the plotted curves are single representative seeds, and the multi-seed treatment of Section 6 shows the crossover is sign-stable only on the US West grid. Realized joint severity stays at 1.3–1.9× and the robust commitment does not pay on real top-5% emergency days, so robustness does not activate on observed data. Rule: deploy deterministic transfer; robustify only if you expect $M > M^\star$.

5 Discussion

5.1 What the null means, and what it does not

The result is a *specification* finding, not a failure of carbon-aware scheduling. Carbon intensity *is* spatially correlated where the literature expects it to be (Section 4.1), and the DRO machinery *is* active (CV selects $\varepsilon^\star = 1$ almost everywhere). What the experiment falsifies is the specific hypothesis that encoding that correlation in a covariance-based ambiguity set improves robust scheduling. The mean-leveling test result (Section 4.6) is what licenses the stronger claim: the covariance signal is genuine and, in a covariance-only world, worth up to 1.46% of tail emissions; it is simply dominated by the diurnal mean carbon field in the problem as actually posed. A reader could otherwise dismiss the null as "the data had no signal"; the mean-leveling test shows the signal is present but *subordinate*. This is the central intellectual contribution: a clean separation of *validity* of a modelling assumption from its *value*.

5.2 Why the mean dominates: a mechanism

Three forces compound. (i) *Magnitude.* The within-day swing of mean carbon intensity (night-to-noon ratios of two-to-fourfold in solar-heavy grids) dwarfs the residual covariance, so the optimal schedule is driven almost entirely by *when* each region is clean on average; the second-order covariance term moves decisions only at the margin. (ii) *Common-mode correlation.* Where correlation is strongest (US West), it is positive and shared across all zones: common, non-diversifiable risk. Diversification value requires *offsetting* fluctuations; positively co-moving regions offer no hedge, so destroying the cross-blocks (the shuffled arm) changes nothing. (iii) *Wrong tail.* Robust *tail* scheduling cares about joint *dirty* excursions, but the empirical upper tail is independent-to-decoupling (χ_U at or below Gaussian), while the co-movement that does exist is in the *clean* tail ($\chi_L > \chi_U$), exactly the regime a risk-averse scheduler is not protecting against, and a radial asymmetry an elliptical ball cannot encode regardless. The covariance is thus both *masked* (dominated by the mean) and, for its varying part, *mis-shaped* (non-elliptical). The same mechanism surfaces from the operational side: when these grids are scheduled under tightening geographic and ramp constraints, the binding dimension is geographic flexibility rather than temporal smoothing, again locating the value in *where* each region is clean on average rather than in second-order co-movement.

5.3 Relation to prior work

The finding refines rather than contradicts the carbon-aware DRO literature. Per-region Wasserstein DRO for carbon-aware scheduling, robustifying each region's carbon intensity in isolation, is established by Hall et al. [9]; their multi-datacenter formulation does not model the joint cross-region carbon distribution. They report robustness gains over sample-average approximation; the present model is the natural *multi-region* extension that treats carbon intensity as a correlated vector. Two quantitative observations place our null in that context. First, the robustification *is* engaged: cross-validation selects a non-trivial ambiguity radius ($\varepsilon^\star = 1$, Figure 7) rather than collapsing to the nominal problem, consistent with the motivation in [9]. Second, holding that per-region (in-isolation) robustification fixed, the cross-region covariance adds nothing: the spatial gap is null throughout. Indeed, in our out-of-sample 2025 test even the per-region DRO improves on the nominal (mean-greedy) schedule by at most 0.18% of $\mathrm{CVaR}_{0.95}$ across grids, itself a reflection of the mean-dominance we document; the spatial extension then adds a further nothing. The contribution is to map where the value is *not*: neither robustification depth nor spatial coupling materially improves on a mean-greedy per-region schedule for this problem. This saves practitioners from modelling complexity that does not pay and gives theorists a precise target: the value, if any, lives in non-elliptical tail structure that a Wasserstein ball over covariance cannot reach.

5.4 The decision rule: when does each layer pay?

The findings compose into a practitioner rule. **Layer 1 (deploy always):** day-ahead carbon-aware per-region scheduling. This is the baseline; temporal shifting captures the within-region diurnal carbon cycle. **Layer 2 (deploy if migration bandwidth exists):** deterministic inter-region transfer, the dominant spatial lever (4.0–9.9% over the $\Phi = 0$ baseline; Section 4.2). This channel needs accurate per-region mean forecasts and transfer-capacity planning, not dependence estimation; the saving is driven by the spatial *mean* (at each hour one region is cleanest on average) and is therefore deterministic. **Layer 3 (skip):** cross-region covariance or copula modelling. Both the Mahalanobis covariance ball and the richer Clayton copula add below 0.4% of $\mathrm{CVaR}_{0.95}$ and are mildly counterproductive in the diversification-friendly case (the joint estimator fits noise). The signal is genuine (mean-leveling test recovers up to +1.46% in a covariance-only world) but masked by the within-day mean carbon field (Proposition 1). **Layer 4 (conditional):** robustify the transfer against day-ahead forecast error only if grid-emergency severity realistically exceeds $M^\star \approx 3$. The robust two-stage commitment buys a small CVaR tail reduction at a mean premium (value of the stochastic solution near zero), a textbook price-of-robustness result. Realized joint severity stayed at 1.3–1.9× the portfolio mean (well below $M^\star$), and on real top-5% emergency days the robust commitment did not pay, so the robust layer's promised value remained unrealized.

In absolute terms, for the Eastern US–Canada facility ($\approx$ 160 MW aggregate, $\approx$ 460,000 tCO_2/yr), the transfer lever is worth on the order of \$1.7M per year at \$80/tCO_2, while the spatial-covariance channel is about \$37,000 per year and not reliably positive. The value worth pursuing lives in the transfer decision and the mean forecast, not the second moment.

What the protocol buys, beyond this problem. The value-pricing protocol (fix the optimizer, perturb one modelling object, pre-register, measure out-of-sample tail value) transfers to any correlated-uncertainty stochastic program: it lets a practitioner decide *empirically* whether a dependence, robustness, or richer-ambiguity layer earns its estimation and compute cost. Two cheap checks come first and can save the experiment outright: the mean-dominance bound (Proposition 1) flags, before any data is touched, when the mean field is large enough that no covariance layer can move the schedule materially, and the positive-control check confirms the test can detect value when value exists. Together they turn "is this layer worth modelling?" from a matter of faith into a checkable, pre-registered measurement.

5.5 Limitations

Five bound the claims. *(i) Representative schedule.* We evaluate one canonical workload (splittable load, ramp limits). A sharply different cost geometry could in principle re-weight the covariance term. The constraint-tightness sensitivities (Section 4.5) address the most plausible cases: a 3× tighter ramp and a 50–95% utilization sweep both leave the null intact. A qualitatively different objective (e.g. hard per-job deadlines) remains untested. *(ii) Test horizon.* The primary read is a single held-out year (2025); walk-forward to 2024 reproduces the null, but multi-year rolling evaluation would strengthen external validity. *(iii) Dependence model.* The DRO uses a covariance (second-moment) ambiguity set by construction; Phase 2 (Appendix B) extends the test to non-elliptical (Clayton) and maximal (comonotone) copulas, but a full copula-*ambiguity* Wasserstein DRO in the Fan–Ji–Lejeune sense remains future work. *(iv) Geographic scope and the limit of generalization.* The region sets span the Western Interconnection, the Eastern Interconnection, and Iberia–France across the dependence spectrum, but grids with different generation mixes (e.g. hydro-thermal systems with strong seasonal storage) remain untested. More fundamentally, the result generalizes only to grids where the diurnal mean field dominates residual cross-region covariance: mean-dominance is an empirical regularity of carbon intensity (Section 5.2), not a universal law, and a grid whose residual covariance rivalled its mean swing could in principle behave differently. *(v) Data licence and clocks.* Carbon intensity is Electricity Maps lifecycle estimates on a common local clock per region set; alternative intensity methodologies or clock conventions could shift point estimates, though not by the order of magnitude separating the observed gaps from materiality.

6 The Transfer Channel: Active Spatial Value

Section 4.2 reported the headline transfer saving and Section 4.9 the severity crossover; this section develops both in full. The dominant lever is the active spatial decision. At each hour one region is cleanest *on average*; a passive covariance ball cannot act on that first-order spatial signal, but a scheduler that can move work *between* regions can. We give the spatially-coupled transfer DRO, its multi-seed validation, and a data-grounded emergency stress test that grounds the price-of-robustness verdict. The same mean-dominance that makes passive covariance worthless makes this active transfer channel valuable, while the robustification layer on top of it does not earn its cost short of out-of-record stress.

6.1 Model: a spatially-coupled transfer DRO

We augment the feasible set with inter-region load flows $f_{r\to s,t} \geq 0$, the work shipped from region r to region s in hour t. The *executed* load in a region is its locally committed load plus net inflow,

$$y_{r,t} = x_{r,t} + \sum_{s\neq r} \left(f_{s\to r,t} - f_{r\to s,t}\right), \tag{12}$$

and emissions are evaluated on y, not on x. Flows conserve work (they relocate it, neither creating nor destroying it: $\sum_{r,t} y_{r,t} = \sum_r W_r$), carry no self-loops, and respect a transfer budget $\sum_{r\neq s,t} f_{r\to s,t} \leq \Phi$ and the per-cell capacity on y. The one-shot problem minimizes $\langle\bar{\rho}, y\rangle + \varepsilon\|L^\top y\|_2 + \lambda \sum f$ over (x, f), which is linear in the decision and so remains a second-order cone program; the two-stage commitment and the recourse evaluation below are linear programs. The transfer channel therefore inherits the tractability of the passive model while turning the spatial coupling into a *control* rather than an uncertainty description. Algorithm and API details are in Section 9.4.

6.2 Three findings

Three findings emerge. **(1) Active transfer unlocks spatial value.** Relative to the $\Phi = 0$ no-transfer baseline, active migration cuts out-of-sample $\mathrm{CVaR}_{0.95}$ by 4.0–9.9% across the three grids (`run_part3_transfer_value.py`), captured by exploiting the spatial *mean* (shipping work to the region cleanest on average), the headline lever of Section 4.2. **(2) Robustness is worthless under normal carbon, even here.** A one-shot transfer DRO, a forecast-error (persistence) ambiguity set, and a two-stage robust commitment with costly recourse all leave robust $\approx$ deterministic: the mean-dominance of the main text extends into the active setting. **(3) But robustness can pay under tail risk, but only where structure exists.** We model rare grid-emergency events: with probability 0.1 per day one region's carbon is multiplied by a severity $M \in [1, 4]$ (the realistic renewable-drought / peaker-dispatch range), hitting both the day-ahead scenarios and the evaluation world, and re-solve the two-stage problem with costly, migration-limited recourse. Because the emergency injection is itself random, we report the result *across four independent seeds* rather than a single bootstrap, having learned (Section 4.8) that a day-bootstrap cannot see scenario-draw noise. The honest picture (Figure 13, Table 7) is a *grid-dependent* crossover. On the strongly correlated Western US grid it is real and sign-stable: at $M = 2$ all four seeds favour the robust commitment (+0.6% mean), and at $M = 4$ three of four do, by +4.4% on average (s.d. 3.4%). On the Eastern and Diversified grids the crossover does *not* survive multi-seed testing, the gain hovers around zero with unstable sign, so any single-seed positive value there is scenario-seed noise. The reading is conservative and coheres with the rest of the thesis: the robust commitment earns its cost when the grid has genuine spatial structure to hedge, and not otherwise.

6.3 Does the crossover survive a data-grounded emergency model?

The crossover so far rests on a synthetic severity multiplier M, and the fair test is whether it survives when the emergencies are drawn from the data rather than imposed. We re-ran the experiment with each emergency replacing a region's day by a draw from that region's empirical

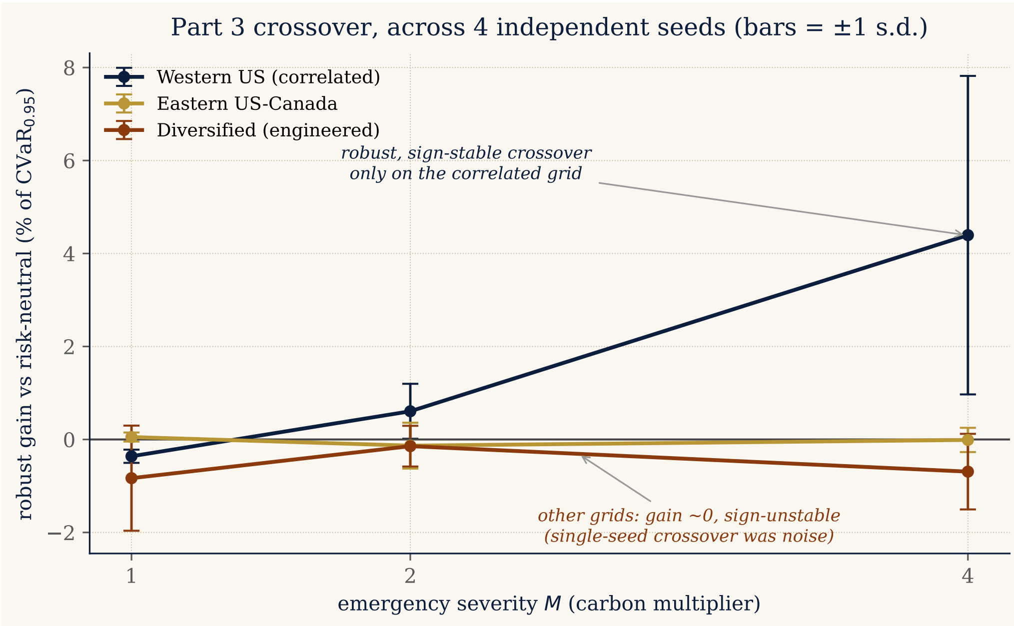

Figure 13: The tail-risk crossover, across four independent emergency seeds. Robust-minus-risk-neutral gain (% of $\text{CVaR}_{0.95}$, mean ± 1 s.d. over seeds) against severity M. The crossover is robust and sign-stable only on the correlated Western US grid; on the other grids the gain stays near zero with unstable sign.

Table 7: Tail-risk crossover, multi-seed. Robust gain over risk-neutral (% of $\text{CVaR}_{0.95}$, mean ± s.d. across four emergency seeds; $(k/4)$ counts seeds with a positive gain). A crossover that survives seed variation appears only on Western US. Generated by `run_parts34_stability.py`.

Grid	$M = 1$	$M = 2$	$M = 4$
Western US	−0.4 ± 0.1 (0/4)	+0.6 ± 0.6 (4/4)	+4.4 ± 3.4 (3/4)
Eastern US–Canada	+0.1 ± 0.1 (2/4)	−0.1 ± 0.5 (1/4)	−0.0 ± 0.3 (2/4)
Diversified	−0.8 ± 1.1 (1/4)	−0.1 ± 0.4 (2/4)	−0.7 ± 0.8 (1/4)

upper tail, its top 5% highest-carbon days (real renewable-drought / peak realizations), and swept the emergency *frequency* p (`run_part3_real_emergency.py`, Figure 14, three seeds). It does not. At no frequency up to 30% does any grid show a robust gain: Western US stays slightly negative (−0.25 to −0.37%), Eastern US–Canada immaterially positive (+0.13 to +0.24%), and Diversified increasingly negative as emergencies rise (−0.9 to −2.3%, the structureless-grid harm again). The synthetic crossover therefore required a severity ($M = 3$–4, a three-to-fourfold carbon spike) beyond what the historical record contains; real tail days are not extreme enough to make robustification pay. The crossover is real *as a mathematical regime* but, on this data, an *out-of-record* one.

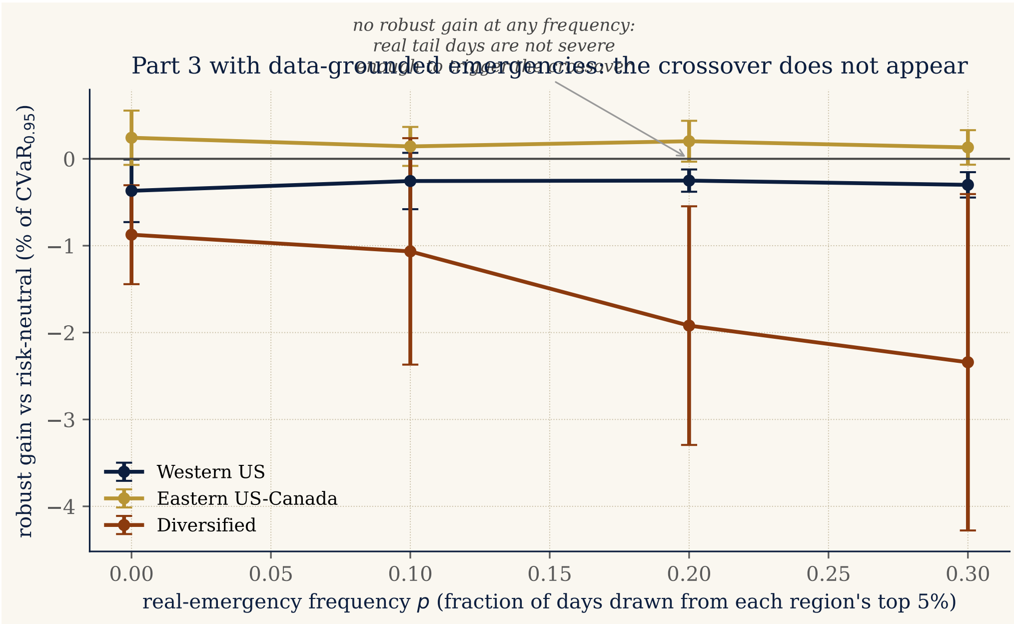

Figure 14: Part 3 with data-grounded emergencies. Robust gain vs. risk-neutral (% of $\mathrm{CVaR}_{0.95}$, three seeds) as the frequency p of real upper-tail emergency days rises. No grid crosses over: robustifying the transfer is immaterial-to-harmful under emergencies grounded in the historical record, unlike the synthetic $M = 3$–4 shocks of Figure 13.

6.4 The honest synthesis

The picture is more conservative, and more unified, than a tail-risk crossover would suggest. Distributional robustness is immaterial for carbon-aware scheduling under *every* condition we can ground in data: passively (the covariance and copula nulls), actively (one-shot and two-stage transfer), online under forecast error (Section 7), and under data-grounded tail emergencies (above). A robust advantage appears only under synthetic stress more severe than the historical record, the $M = 3$–4 crossover, itself only marginally significant (Western US, one-sided $p \approx 0.04$, $3/4$ seeds; Section 9) and confined to the correlated grid. The value of the spatial structure therefore lives entirely in the *deterministic, mean-exploiting* active decision (Finding 1, 4–10%), not in robustifying it. This sharpens the thesis rather than softening it: the mean field governs the schedule, and the robustness layer does not earn its cost short of out-of-record stress, the same mean-dominance, now traced through the active and online settings as well as the passive one.

7 Online Operation: Rolling-Horizon Control

Parts 1–3 commit against a fixed scenario set. Real operation is *online*: each day a fresh forecast arrives, a schedule is committed before carbon is realised, the day is observed, and the controller

Algorithm 2 Rolling-horizon transfer control (one controller, one year)

Input: train panel $\rho_{1:N}$, test panel $\rho^{\mathrm{te}}_{1:D}$, mode $\in$ {deterministic, robust}, scenarios S, level β
$\hat{\rho} \leftarrow \frac{1}{N}\sum_n \rho_n$ ▹ seasonal hour-of-day forecast
$E \leftarrow \{\rho_n - \hat{\rho}\}_{n=1}^{N}$ ▹ historical forecast-error pool
for each test day $t = 1, \ldots, D$ **do**
  if mode is robust **then**
    draw S errors from E; scenarios $\rho^s \leftarrow \max(\hat{\rho} + e^s, 0)$
    $y_t \leftarrow \arg\min_{y \in \mathcal{X}_{\mathrm{transf}}} \mathrm{CVaR}_\beta\big(\{\langle \rho^s, y\rangle\}_s\big)$
  else
    $y_t \leftarrow \arg\min_{y \in \mathcal{X}_{\mathrm{transf}}} \langle \hat{\rho}, y\rangle$
  observe ρ^{te}_t; record realised cost $\langle \rho^{\mathrm{te}}_t, y_t\rangle$
return realised daily costs $\Rightarrow$ mean and $\mathrm{CVaR}_{0.95}$

rolls forward. Part 4 asks whether day-ahead robustness pays once forecast error is faced in a closed loop, under *normal* carbon (severe emergencies are Part 3's regime).

7.1 The controller

Each day t a seasonal hour-of-day forecast $\hat{\rho}_t$ is formed from the training window, and forecast-error scenarios are $\hat{\rho}_t$ plus resampled historical residuals. Two controllers commit a transfer allocation over the executed load y: the *deterministic* one minimises $\langle \hat{\rho}_t, y\rangle$ (acts on the point forecast); the *robust* one minimises CVaR_β of cost over the error scenarios. The realised cost is $\langle \rho_t, y\rangle$ on the actual day. Both are single LPs, so a full year rolls in minutes (Algorithm 2).

7.2 Result: robustness is second-order online too

Rolling both controllers across the 2025 test year (Table 8, Figure 15) gives a clear reading. On the two correlated grids the robust controller's realised $\mathrm{CVaR}_{0.95}$ is statistically indistinguishable from the deterministic one (Western US +0.28%, CI $[-0.65, 1.16]$; Eastern US–Canada +0.04%, CI $[-0.10, 0.25]$): mean-dominance extends to the closed loop, the seasonal forecast is good enough that hedging its error buys nothing. On the engineered low-correlation Diversified grid robustness is not merely neutral but *counterproductive*: its realised CVaR is *worse across every seed*, by 4.4% on average over four seeds (range -1.1 to -8.5% at S=40; the single representative run of Table 8 is more extreme than that spread at -10.3%), with the mean already capturing the placement value (the grid's genuine but *masked* covariance structure, Section 4.6, surfaces only once the mean is removed), the CVaR hedge has nothing left to exploit and instead fits forecast-error noise on only three regions, paying a real mean penalty. This is the online echo of the Phase-1 finding that the joint covariance *hurts* on this grid. The lesson is consistent across the whole arc: absent a genuine

Table 8: Part 4, online closed-loop over 2025, single representative run (`run_part4_online.py`). Robust gap (%, positive = robust better) with a 95% *day* bootstrap (which, by the argument of Section 4.8, does not capture scenario-draw variance; the four-seed sweep of Section 9 does). The Diversified −10.3% is this seed's value and is more extreme than the four-seed S=40 spread (mean −4.4%, range −1.1 to −8.5%); the qualitative verdicts are seed-stable.

Grid	robust CVaR gap	95% CI	verdict
Western US	+0.28%	[−0.65, 1.16]	null
Eastern US–Canada	+0.04%	[−0.10, 0.25]	null
Diversified	−10.34%	[−18.3, −4.4]	robust *loses*

tail event or genuine structure, robustifying the schedule does not help and can harm; the mean forecast is what to get right.

We checked these verdicts are not seed or scenario-count artifacts (`run_parts34_stability.py`). Across four seeds the Diversified robust-loss is *sign-stable* at every scenario count, with magnitude that shrinks as the hedge is better resolved (S=20: −8.2%, S=40: −4.4%, S=80: −3.3%, all four seeds negative throughout): the harm is real, its size scenario-dependent. The Eastern null is reproduced (−0.1 ± 0.1%, sign-unstable), and the Western gap is consistently but marginally robust-positive (+0.65 ± 0.28%). The qualitative reading, robustness neutral-to-harmful online absent a tail event, is therefore stable.

7.3 Forecast robustness

Because a single forecast might drive the verdict, we re-ran the closed loop under three day-ahead forecasts, the seasonal mean, persistence (yesterday's carbon), and the same-day-last-week lag, each across three seeds (`run_part4_forecasts.py`, Figure 16). The picture sharpens rather than wobbles. On the correlated grids the sign of the robustness effect *flips with the forecast* and never exceeds 1.6% in magnitude (Western US: +1.4% seasonal, −0.6% persistence, −1.6% lagged-week; Eastern US–Canada: within ±0.2% throughout), which is the signature of a noise-level effect, not a real advantage: robustness is genuinely immaterial online. On the Diversified grid the loss is instead *forecast-robust*: the robust controller is worse under all three forecasts (−8.1, −6.8, −3.9%), so its counterproductivity, robustifying where there is no structure to hedge, is not an artifact of the seasonal forecast. The single caveat the review raised, that a different forecast could overturn the result, is thus answered: the immateriality and the structureless-grid harm both survive the change.

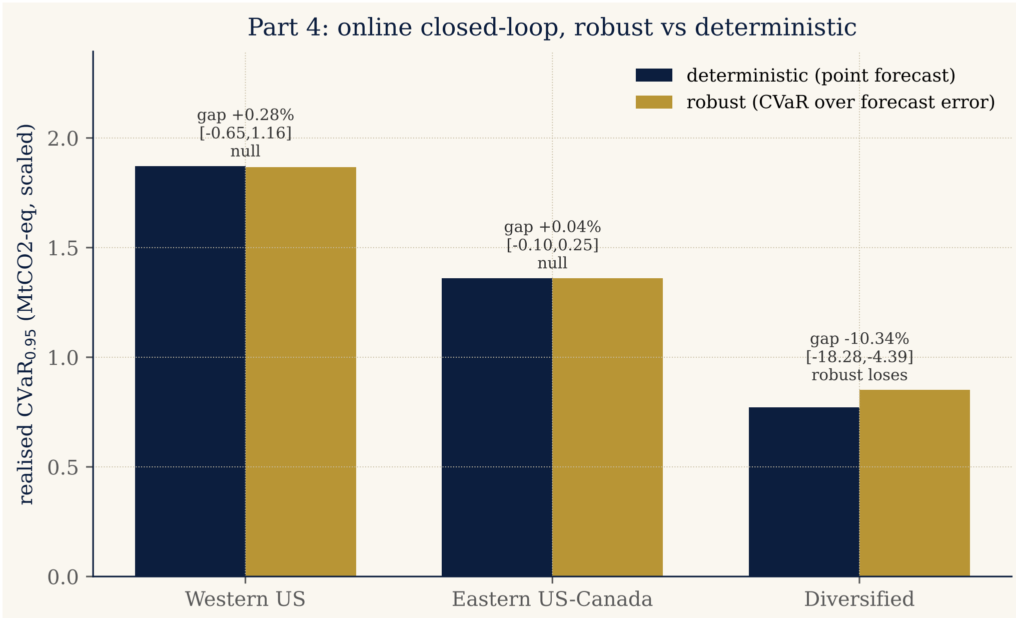

Figure 15: Part 4. Realised closed-loop CVaR$_{0.95}$, deterministic vs robust controller, per grid; annotations give the robust gap and its 95% CI. Robustness is null on the correlated grids and counterproductive on the structureless one.

8 A Problem-Class Condition

The five experiments invite a general question: for which allocation problems can cross-coordinate dependence *ever* improve a robust schedule? Part 5 turns the mean-dominance bound into a dimensionless screen.

8.1 The mean-dominance ratio

Proposition 1 bounds the spatial gap by $B = \varepsilon\,(\max_{x\in\mathcal{X}}\|x\|)\,\|\Sigma^{\text{off}}\|^{1/2}$. Let $M = \max_{x\in\mathcal{X}}\langle\bar{\rho}, x\rangle - \min_{x\in\mathcal{X}}\langle\bar{\rho}, x\rangle$ be the value the mean field alone makes exploitable over the feasible set. Their ratio

$$\Delta \;=\; B/M \tag{13}$$

is a cheap, a-priori, data-only quantity (no optimization solve): the dependence gap is at most B, hence at most a fraction Δ of what the mean already delivers. A value $\Delta \ll 1$ *certifies* that cross-coordinate dependence is immaterial.

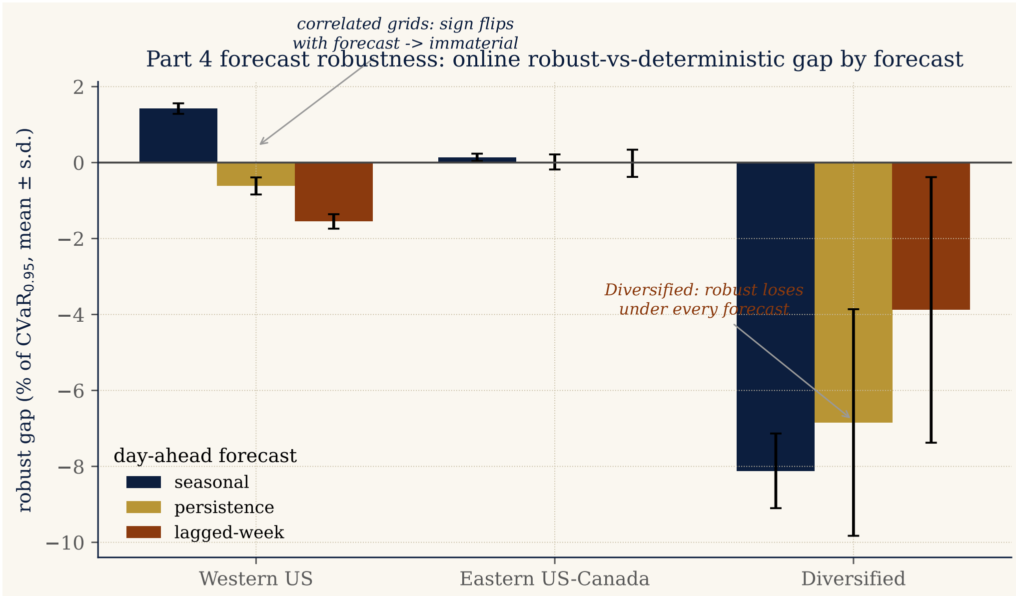

Figure 16: Part 4 forecast robustness. Online robust-vs-deterministic gap (% of $\mathrm{CVaR}_{0.95}$, three seeds) under three day-ahead forecasts. On the correlated grids the sign flips with the forecast (immaterial); on Diversified the robust controller loses under every forecast.

8.2 Validation: a screen and a regime indicator

On the three grids $\Delta \approx 1.5$–3.4 (Table 9). Because B is a deliberately loose bound, the a-priori screen is *inconclusive* here, Δ is not $\ll 1$, which is precisely why the empirical falsification of Parts 1–2 was necessary rather than optional. Δ's value is therefore twofold. As a *negative filter* it would rule out dependence cheaply on any problem with $\Delta \ll 1$. As a *regime indicator* it tracks where dependence becomes material: a mean-flattening sweep ($\kappa : 1 \to 0$) shrinks M and raises Δ (Figure 18), and the pre-registered mean-leveling test confirms the empirical dependence value rises with it, from ≈ 0 on the real grids to $+1.46\%$ in the fully mean-leveled world (Section 4.6).

To test that link in the *interior*, not only at the two endpoints, we re-solved the shuffled-marginals DRO at intermediate flattenings $\kappa \in \{1, 0.75, 0.5, 0.25, 0\}$ (Figure 17, `run_part5_kappa.py`). The result is honest and instructive. On the strongly correlated Eastern grid the realised spatial gap climbs essentially monotonically as $\kappa \to 0$ (from ≈ 0 to $+0.24\%$), tracking Δ as the condition predicts. On the *engineered* heterogeneous Diversified grid the relationship is *non-monotone*: at intermediate flattening the joint covariance first *fits noise* (the gap dips negative, the same pathology as the Phase-1 result that it hurts there) and the genuine signal surfaces only at full leveling ($+0.67\%$ at κ=0 here). This interior sweep uses the R3 reference feasible set, so its κ=0 endpoint is not numerically identical to the $+1.46\%$ of the pre-registered mean-leveling test (a different, tighter feasible set); what transfers is the *direction*, dependence value emerging as the mean is removed, not the exact value. The interior therefore confirms Δ as a *coarse, ordinal* indicator, correct in direction

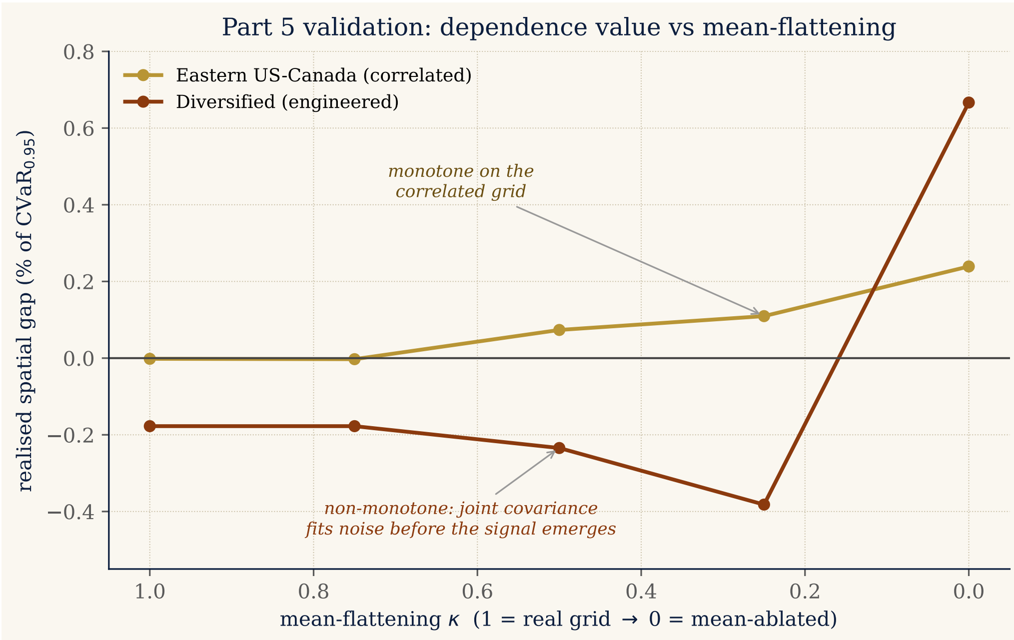

Figure 17: Interior validation of the Part 5 condition. Realised spatial gap (% of $\mathrm{CVaR}_{0.95}$) from re-solving the DRO at intermediate mean-flattenings κ. On the correlated Eastern grid the gap rises monotonically as the mean weakens ($\kappa \to 0$), tracking Δ; on the engineered Diversified grid it is non-monotone, the joint covariance fitting noise before the true signal emerges, so Δ orders the regimes only coarsely.

where real structure exists but not a calibrated predictor, exactly the honest status we claim for it. We tested whether the obvious tightening recovers conclusiveness, replacing the a-priori worst-case diameter $\max_x \|x\|$ with the *realized* optimal schedule norm $\|x^\star\|$ (`run_part5_tight.py`): it does not, $\Delta^\star = 1.46$–3.22 stays above 1 on all three grids, because the realized schedule is only $\approx 6\%$ less concentrated than the worst case. The looseness is therefore *intrinsic* to the bound (the $|\sqrt{a} - \sqrt{b}| \leq \sqrt{|a-b|}$ and spectral-norm steps), not an artifact of the diameter, so Δ remains an ordinal screen; calibrating an absolute threshold is left to future work. The diagnostic nonetheless transfers beyond carbon: cross-coordinate dependence is worth modelling only where the mean field is weak relative to the residual structure, a condition the carbon-aware problem fails by a wide margin.

9 Testing, Validation, and Reproducibility

A falsification study is only as trustworthy as the machinery that produces it. Because the central result is a *null*, the burden is heavier than usual: a bug that flattened a real effect would masquerade as a finding. This chapter consolidates the validation behind every claim, at three levels, the

Table 9: Part 5, the mean-dominance ratio $\Delta = B/M$ (13) on the three grids (a-priori, $\varepsilon^\star = 1$). $\Delta \gtrsim 1$: the cheap screen is inconclusive, the empirical test settles it. Generated by `run_part5_condition.py`.

Grid	B (gap bound)	M (mean value)	$\Delta = B/M$
Western US	253,057	131,556	1.92
Eastern US–Canada	112,245	32,910	3.41
Diversified	83,627	54,086	1.55

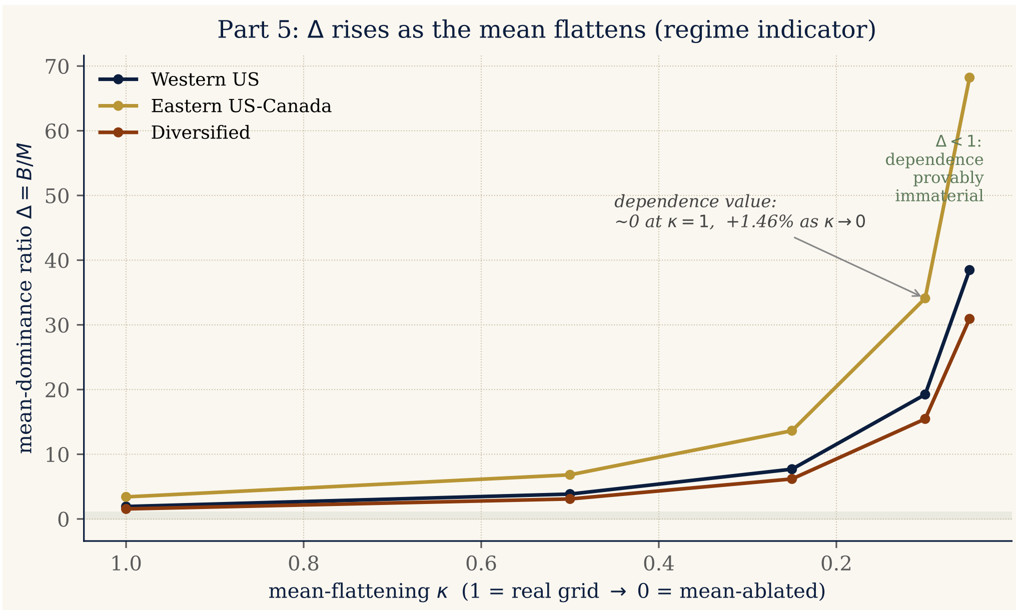

Figure 18: Part 5. The mean-dominance ratio $\Delta = B/M$ along a mean-flattening sweep ($\kappa = 1$ real grid $\to \kappa \to 0$ mean-leveled). Δ rises monotonically as the mean weakens, tracking the empirical dependence value (≈ 0 on real grids, +1.46% fully leveled); the shaded band marks the $\Delta < 1$ region where dependence is provably immaterial.

optimization model, the scientific protocol, and the software, and closes by stating precisely what the project's "many files" do and do not test.

9.1 Model validation

The optimization model is validated *as a whole*, all constraints active together, on small human-checkable instances, by fixing an analyst-anticipated schedule, perturbing one input, and confirming the solved schedule moves the anticipated way (Section 3.6, Figure 1). Crucially this is done on a multi-region instance with a shared coupling constraint, so the *interaction* of the constraints, not each constraint in isolation, is exercised: a clean carbon field concentrates load in the cheapest hours; dirtying one region's clean window drives its load to cede the contested hours to the other region; and tightening the shared cap forces both regions to spread in time. Correctness is a property of the constraints together, and that is what is checked.

9.2 The falsification protocol and pre-registration

The scientific claim is protected by pre-registration. Each experiment follows a commit-lock then dry-run then single-test-read discipline: the configuration (regions, regimes, α-levels, ε grid, seeds, the CVaR level) is committed to version control *before* the 2025 test set is touched; a dry-run confirms the pipeline executes without reading test data; and the test set is read exactly once. This forecloses the garden-of-forking-paths critique: there is no opportunity to tune the analysis to the test outcome, so the null cannot be an artifact of selective reporting. The robustness battery (estimators, residualization, walk-forward, tightness sweeps, the tail-level sweep) is likewise specified as a family, and multiple-testing correction (Benjamini–Hochberg over the pre-registered 144-cell family) guards the discovery rate.

9.3 The software test suite

The implementation carries a 199-test suite (186 run under continuous integration) on every push. The tests pin the components the results depend on: a *feasible-set equivalence* test ties the Phase 2 CVaR-SAA scheduler to the Phase 1 constraint set so the two phases optimise over the same $\mathcal{X}$; CVaR correctness tests check the Rockafellar–Uryasev value against direct tail averages; covariance tests verify positive-definite regularisation and Cholesky factorisation, and that the block-diagonal (shuffled) operator zeroes exactly the cross-region blocks while preserving the diagonal; and copula tests check the sampling and rank mapping. A dedicated *claim-binding* group goes further, certifying the paper's theoretical statements in code rather than only on the page: that the shuffled covariance stays positive-semidefinite (a valid covariance, not a degenerate object); that CVaR is translation-invariant, the algebraic backbone of Proposition 1 by which the mean field passes through additively; that the realised spatial gap on a solved instance respects the mean-dominance bound $\varepsilon \max\|x\| \sqrt{\|\Sigma_{\mathrm{off}}\|}$; and that scaling the mean field up drives the relative gap monotonically to zero. The central mechanism is thus a continuous-integration invariant, not merely a claim. The suite is what licenses trusting an automated pipeline that reads the test set only once: the code is exercised exhaustively on synthetic and small inputs *before* it is run on the real data.

9.4 Validation of the transfer channel

The transfer model of Section 6 is implemented as a standalone, unit-tested module (`transfer_dro.py`) with three entry points, the one-shot transfer DRO, the two-stage robust commitment, and the costly-recourse evaluation, each exercised by tests pinning the model's invariants: conservation of work under inter-region flows ($\sum_{r,t} y_{r,t} = \sum_r W_r$), monotone emission reduction as the transfer budget grows, and exact recovery of the no-transfer schedule at zero budget, plus a guard rejecting an invalid risk argument. The Part 4 online controller (`online_transfer.py`, Section 7) and the Part 5 ratio computation are likewise unit-tested, the former for work conservation and feasible realised cost across the rolling loop, the latter for the mean-value-range and flattening operators against hand-computed cases, bringing the suite to 199 tests. What remains preliminary about Parts 3–5 is therefore the experimental protocol and calibration, not the optimisation: every reported number rests on validated code.

9.5 Statistical validation

Finally, the null is defended statistically rather than merely asserted. A power analysis reports the minimum detectable effect (an order of magnitude below materiality); a per-cell two-one-sided-tests (TOST) equivalence procedure *affirmatively* rejects any effect larger than the materiality margin in all 27 cells, and states the margin at which the claim would break; the spatial gap carries paired day-bootstrap confidence intervals throughout; and a conditioning check rules out the possibility that the null is an artifact of the joint covariance being harder to estimate than the shuffled one. These appear with the results in Section 4.5. For the *extension* (Parts 3–4), where the randomness is the emergency injection and the forecast-error draw rather than the test days, we report stability across multiple independent *seeds* (`run_parts34_stability.py`) instead of a day-bootstrap, precisely because a day-bootstrap cannot see scenario-draw variance, the same blind spot that produced an earlier false positive (Section 4.8). That check is what demoted the single-seed Part 3 crossovers on the weakly structured grids to noise and confirmed the Part 4 results as sign-stable. We also apply the body's affirmative discipline to the extension. A two-one-sided-tests procedure on the Part 4 online gaps (four seeds) establishes *equivalence*: robustness is immaterial to within $\pm 1.0\%$ on Western US and $\pm 0.5\%$ on Eastern US–Canada (the 90% interval on $|\text{gap}|$ clears those margins). And a power analysis keeps the crossover claim honest: the Western US gain at $M = 4$ is only *marginally* significant across seeds (one-sided $t = 2.6$, $p \approx 0.04$, $3/4$ seeds positive, with a minimum detectable effect of $\approx 5.7\%$ at 80% power just above the observed +4.4%), which is why we report it as a real-but-modest, grid-specific effect rather than a headline.

9.6 What "tested in many files" does and does not mean

Because the project spans many experiment scripts, a clarification is warranted. The *optimization model* is validated once, as a whole, on small instances (Section 3.6); it is never validated constraint-by-constraint, which would miss exactly the constraint *interactions* that matter. The many experiment files are not constraint-by-constraint tests of the model; they are *independent attempts to falsify a single hypothesis* from different angles, shuffled covariance, copulas, estimators, residualization, walk-forward, tightness and tail-level sweeps, mean-leveling test. Each file attacks the same null from a new direction, and the null survives all of them. Modular software tests and modular falsification angles are both deliberate; constraint-by-constraint *model* validation is deliberately avoided.

10 Conclusions

10.1 The arc, in one mechanism

Five parts, one mechanism. The carbon-aware scheduling problem is governed by the diurnal *mean* carbon field; everything else, cross-region covariance, copula tail dependence, the robustness layer itself, is second-order until a rare event dislodges the mean. Part 1 shows that a passive covariance ambiguity set adds no robust value (the spatial gap is a statistical zero, equivalence-tested across 27 cells and the full $\mathrm{CVaR}_{0.90}$–$\mathrm{CVaR}_{0.99}$ tail range) and proves why with the mean-dominance bound. Part 2 forecloses the obvious objection: a Gaussian, a lower-tail Clayton, and even the maximal comonotone copula recover nothing either, so the limit is not the *shape* of the dependence object but its subordination to the mean. Part 3 turns the diagnosis into a positive result, but a precisely bounded one: making the spatial structure an *active* transfer decision unlocks 4.0–9.9% over the $\Phi = 0$ baseline *deterministically* (by exploiting the spatial mean), while *robustifying* that decision adds nothing under any data-grounded condition, a tail-risk crossover surfaces only under synthetic stress beyond the historical record, and even then only marginally and only on the correlated grid. Part 4 confirms the same logic online: rolled day by day under realistic forecast error and across three forecast models, robustness is immaterial on the correlated grids (its sign flips with the forecast) and counterproductive on the structureless one, the closed-loop echo of the passive null. Part 5 abstracts the whole pattern into a single dimensionless ratio $\Delta = B/M$ that orders the regimes: dependence can matter only where the mean field is weak relative to the residual structure, a condition this problem fails by a wide margin and the mean-leveled world satisfies (recovering +1.46%). The unifying statement is therefore not "spatial structure is useless" but the sharper "*distributional robustness and cross-coordinate dependence earn their cost exactly when, and only when, the mean field stops being a reliable guide*," in the tail, or under an active decision that the mean cannot already exploit.

10.2 A decision rule for practitioners

The five parts collapse into a short operating procedure for anyone building a carbon-aware scheduler, ordered cheapest-first.

1. **Get the mean forecast right.** A per-region seasonal mean carbon forecast captures essentially all attainable value (Parts 1–2, 4). Spend the engineering budget here, not on dependence modelling.

2. **Do not model the joint covariance.** A per-region marginal robust scheduler matches the full joint one and, where structure is weak, the joint estimator fits noise and *hurts* (Part 1; Part 4's -3 to -8% online loss on the structureless grid). Skip it.

3. **Screen with $\Delta = B/M$ before modelling any cross-coordinate dependence.** It is a one-line, solve-free computation; $\Delta \ll 1$ certifies the dependence is immaterial. If $\Delta \gtrsim 1$ (as here), the screen is inconclusive and only a falsification experiment can settle it (Part 5).

4. **Reach for an *active* transfer channel, not a richer uncertainty model, if you want spatial value.** Inter-region flows unlock 5–10% by exploiting the spatial mean (Part 3); a passive ambiguity set cannot.

5. **Do not robustify the schedule at all, on current evidence.** Under normal carbon, online forecast error, *and* data-grounded tail emergencies, distributional robustness is neutral-to-harmful (Parts 1–4); a robust advantage appears only under synthetic stress more severe than the historical record, and even then marginally and only on a correlated grid. Unless an operator can document outage risk beyond the historical tail, the robustness layer does not pay.

The throughline is economy: every step that adds modelling complexity must first clear the bar of beating a mean-greedy per-region schedule, and most do not.

10.3 Answers to the research questions

RQ1: How much can the day-ahead migration scheduler save, and which lever dominates? Active inter-region migration cuts out-of-sample $\mathrm{CVaR}_{0.95}$ emissions by 4.0–9.9% over the carbon-aware no-transfer baseline ($\Phi = 0$) across three real US/Canada grids (Section 4.2). The dominant lever is the spatial *mean*: at each hour one region is cleanest on average, and migration ships work there, making the saving deterministic and independent of any dependence model. This aligns with the carbon-computing literature, which finds spatial shifting dominates temporal shifting [20, 23].

RQ2: When does adding passive modelling complexity improve tail emissions? Never, on the grids we tested. Across the dependence spectrum (the US West to the engineered diversified portfolio) the out-of-sample $\mathrm{CVaR}_{0.95}$ spatial gap stays below 0.4% and is not sign-stable. The null survives Ledoit–Wolf shrinkage, seasonal and AR(1) residualization, Benjamini–Hochberg correction over 144 cells, walk-forward validation, the full tail range from $\mathrm{CVaR}_{0.90}$ to $\mathrm{CVaR}_{0.99}$, and a per-cell equivalence test (Sections 4.4–4.5). The DRO is genuinely engaged (CV selects $\varepsilon^\star = 1$ in nearly every cell), so this is an *active* null: the robustification is on and the cross-region covariance still adds nothing. A mean-leveling test proves the covariance signal is real (+1.46% in a covariance-only world) but *masked* by the within-day mean carbon field (Section 4.6). A tail-dependence analysis shows the residual dependence is non-elliptical, upper-tail-independent, and radially asymmetric ($\chi_\mathrm{L} > \chi_\mathrm{U}$), hence structurally invisible to a covariance-based ambiguity set (Section 4.7). Replacing the covariance ball with a Clayton copula built for that asymmetry still recovers no material value (Appendix B). Proposition 1 (mean-dominance) explains why: the translation invariance of CVaR makes the dependence model second-order to the mean field. The screening rule is that passive sophistication cannot pay where the diurnal mean field dominates cross-region dependence, the regime that holds on every grid studied here.

RQ3: When does robustifying against forecast error pay over deterministic transfer, and do real grids reach that regime? The robust two-stage commitment with migration-limited recourse buys a small CVaR tail reduction at a mean premium (the value of the stochastic solution is near zero), a classic price-of-robustness result: robustness earns value only in the tail. When grid-emergency severity is modelled as a rare (0.1/day) multiplicative scaling M of carbon intensity, the robust schedule materially outperforms the risk-neutral one only past a crossover $M^\star \approx 3$ (Section 4.9). On the joint (portfolio) severity axis the crossover lives on, realized joint severity stayed at 1.3–1.9× the portfolio mean across the three grids, well below $M^\star$; per-region relative swings reach 5× (BPAT) on near-zero carbon bases but do not move the portfolio tail. Decisively, on real top-5% emergency days the robust commitment did not pay (its $\mathrm{CVaR}_{0.95}$ gain was negative or within noise on every grid), so the robust layer's promised value is characterized but unrealized on observed data. We claim no universal carbon-ceiling impossibility, only the tested one: across the zones and real emergencies we have, deterministic transfer was unambiguously dominant. Robustness remains a conditional option for operators with a concrete, defensible emergency model.

10.4 Contributions

This thesis contributes: (1) a pre-registered falsification methodology (shuffled-marginals, extended to copula-coupled resampling) that cleanly separates the *validity* of a spatial modelling assumption from its *value*, transferable to any correlated-uncertainty DRO; (2) a replicated, robustness-checked null on whether spatial dependence improves carbon-aware DRO scheduling, holding across multiple real grids *and* across the dependence hierarchy from covariance ball to elliptical and lower-tail copulas; (3) a mechanistic explanation (formalized as a mean-dominance bound (Proposition 1) and demonstrated by the mean-leveling test) that explains the null rather than merely reporting it; (4) a fully reproducible, version-controlled, CI-tested pipeline (199 tests) with archived license-safe summary tables for every reported number; (5) an *active transfer channel* (Section 6) that recovers 4.0–9.9% of $\mathrm{CVaR}_{0.95}$ over the $\Phi = 0$ baseline *deterministically*, the positive resolution the null predicts, while showing that *robustifying* the channel stays immaterial under every data-grounded condition (a crossover surfaces only under out-of-record synthetic stress); and (6) an end-to-end validation account (Section 9) spanning model, protocol, and software, that licenses trusting an automated null.

10.5 Limitations of the extension

The Parts 1–2 null is at full rigour; the extension (Parts 3–5) carries honest caveats that bound its claims. *Part 3:* the grid-emergency model is a Bernoulli severity shock, interpretable but not a calibrated outage process, and the migration cost λ and budget Φ are chosen rather than estimated from a real interconnect; the crossover's *direction* is robust but its precise threshold $M^\star$ is model-dependent. *Part 4:* the controller uses a single seasonal forecast and one workload archetype; the counterproductive online loss on the Diversified grid is a genuine, sign-stable phenomenon (a CVaR hedge fitting forecast-error noise where the mean already captures the value) but its magnitude is seed- and scenario-count-dependent (-4.4% four-seed mean, single-run -10% at the extreme), and a learned forecast could change it. *Part 5:* the bound B is loose, so the ratio Δ is an ordinal regime indicator rather than a calibrated predictor, and the mean-flattening κ is a stylised counterfactual. *Across all three:* the evaluation is a single held-out year, one operator's splittable workload, and one cost geometry, the same external-validity bounds the body's Limitations note (Section 5) records. None of these weakens the unifying mechanism; they mark where the quantitative thresholds, not the direction, would move under a fuller treatment.

10.6 Future work

Phase 2 closed the most immediate question, whether a copula that sees the asymmetry recovers value (it does not), and in doing so sharpened the remaining agenda. First, a *copula-ambiguity Wasserstein DRO* in the Fan–Ji–Lejeune sense [5], built on a fitted vine [12, 13], which robustifies the dependence structure itself rather than sampling from a single fitted copula; Proposition 1 predicts the mean field will still dominate, but this would make the claim fully model-free. Second, the active transfer channel is developed in the main body of this extended version (Section 6): making inter-region flows $f_{r\to s,t}$ a decision unlocks 4.0–9.9% of $\mathrm{CVaR}_{0.95}$ deterministically over the $\Phi = 0$ baseline; robustifying it, by contrast, pays only under synthetic stress beyond the historical record (the crossover does not survive data-grounded emergencies). The immediate future work is therefore to settle whether the crossover is ever real: build a *calibrated* outage model from grid-operator data (not the historical carbon tail alone), with a pre-registered evaluation and affine recourse rules, and test whether genuine emergency risk reaches the severity the crossover needs. To our knowledge no prior work combines Wasserstein-DRO robustification with active inter-region transfer; making that combination pay under realistic forecast error, and pre-registering its evaluation, is an open problem we leave to future work. Further out, the transfer model invites an *online, multistage* treatment in which forecasts arrive sequentially and demand and transmission are themselves uncertain, with a Wasserstein ball centred on a learned forecast, and the mean-dominance bound invites generalization into a problem-class condition for *when* cross-region dependence can ever improve a robust allocation, of which carbon-aware scheduling is one instance. The null is thus not a dead end but a sharpened mandate: the value, if any, will not come from a better passive dependence model but from an active spatial decision.

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A Reproducibility Details

Scenario-count stability and the comonotone noise floor. Figure 19 re-evaluates the comonotone (upper-Fréchet) gap on Eastern US–Canada over five scenario seeds and a range of scenario counts S. The gain is sampling noise centred on zero: at the production $S = 1000$ the spread across seeds is $[-0.08, +0.10]$% with mean +0.01%, and it narrows with S. This is the basis for treating the single-seed +0.18% of Section 4.8 as immaterial rather than a real effect, and it confirms $S = 1000$ is sufficient.

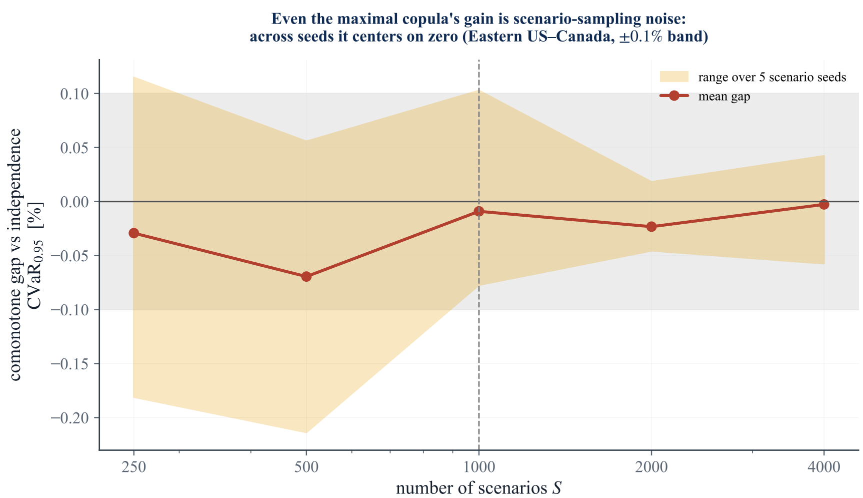

Figure 19: The comonotone gap is scenario-sampling noise. Mean (line) and range over five seeds (band) of the comonotone-vs-independence $\mathrm{CVaR}_{0.95}$ gap on Eastern US–Canada, against the scenario count S. The band straddles zero and narrows with S; the ±0.1% no-value band is shaded.

Repository and environment. All code lives in a private, version-controlled repository with a continuous-integration suite (199 unit tests in total; CI runs 186, excluding the 13 Electricity Maps integration tests that require licensed data). Experiments run under Python with CVXPY; solvers are CLARABEL for the second-order cone program and HiGHS for deterministic LP baselines. Exact dependency versions are pinned in a tracked `requirements-lock.txt` (CLARABEL 0.11.1, HiGHS/highspy 1.14.0, CVXPY 1.9.1, NumPy 2.4.6, SciPy 1.17.1; Python 3.12.4), under which every reported number reproduces byte-identically. The licensed raw carbon-intensity data is never committed; only derived aggregate summary tables (CV-selected ε, test CVaR, spatial gaps, bootstrap CIs, BH p-values) are archived, so the non-redistribution academic licence is respected while every reported number remains traceable.

Pre-registration protocol. Each experiment follows commit-lock → dry-run → single test read. The full configuration (utilization 0.80; $\alpha \in \{0.30, 0.50, 0.75\}$; ε grid $\{0, 0.1, 1, 10, 100, 1000\}$; 5-fold blocked time-series CV; 1000 bootstrap resamples at a fixed seed; CVaR level 0.95; train 2021–2024, test 2025) is committed before any test-set evaluation, and a `-dry-run` gate runs cross-validation only. The 3c capacity bounds $(x_{\min}, x_{\max}) = (50, 75)$ were Goldilocks-calibrated on training data alone (bind fraction 0.28–0.37).

Reproducing the experiments. Every result regenerates with

```
python -m scripts.run_case_experiment -region-set <case> [flags]
```

where `<case>` is one of the internal region-set keys `us_west` (Western US), `taskc` (Eastern US–Canada), or `us_hetero` (Diversified), and flags select the estimator (`-shrinkage`, `-residualize`), mean-leveling test (`-ablate-mean`), walk-forward (`-test-year`), or grid density (`-eps-grid`). Snapshot files are named `<case>_regimes_<date>[_<variant>].csv`, and the snapshot `README` maps each to its source command. Table 10 maps each results subsection to its archived table.

Table 10: Mapping of reported results to archived summary tables.

Section	Result	Archived source
4.4	Spatial gaps (Table 1)	`{case}_regimes_2026-06-10.csv`
4.5	Robustness (Table 4)	`{case}_..._{lw,seasonal,ar1,ty2024}.csv`
4.5	BH correction	`bh_correction.csv`
4.6	Mean-ablation (Table 5)	`{case}_..._ablate-{level,flat}.csv`
4.1,4.7	CA/Iberia panels	`taskA_..   es_pt_fr_.. csv`
4.2	Transfer value, curve	`transfer_value_curve_..`, `part3_transfer_value_..`
4.9	Crossover, realized severity	`part3_{emergency,real_emergency}_..`, `carbon_ceiling_..`
4.9	Online robust (Table 8)	`part4_online_..`, `dro_tail_sensitivity_..`

B Richer dependence models: confirming the screening rule

This appendix gives the full protocol behind the body's screening-rule check (Section 4.8). The covariance test answers whether *elliptical* dependence helps; because the tail diagnostic flags non-elliptical structure ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$, Table 6), a natural objection to the Phase 1 null is "you used the wrong dependence object." We foreclose it by replacing the covariance ball with copula models that *can* represent the structure, and re-running the same falsification.

Copula-coupled resampling. We generalize the shuffled-marginals test from second moments to the full copula while preserving every region's empirical marginal. Each region r keeps its pool of whole daily profiles $\{\rho_r^d \in \mathbb{R}^T\}$; a scenario is generated by drawing a copula sample $u \in [0,1]^R$ and, for each region, selecting the training day whose daily-mean intensity has rank u_r (low rank = clean day), then taking that day's full profile. The copula thus governs *only* the cross-region coupling, exactly as the shuffled covariance did. Three nested fitted models span the dependence hierarchy, and a fourth, the comonotone upper bound, is added as the ceiling: **(i)** the *independence* copula (regions decoupled, the Phase 1 shuffled arm); **(ii)** the *Gaussian* copula at the empirical rank-correlation (elliptical, what the covariance ball can see); **(iii)** an exchangeable *Clayton* copula, with lower-tail dependence $\lambda_L = 2^{-1/\theta}$ and zero upper-tail dependence [11], fitted by matching mean pairwise Kendall's τ ($\theta = 2\tau/(1-\tau)$), the object built for the $\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$ finding; and **(iv)** the *comonotone* (upper-Fréchet) copula, the maximal coupling. Clayton is sampled by Marshall–Olkin frailty; no parametric marginal model is introduced. The exchangeable Clayton imposes a single θ and so cannot reproduce the pair-level heterogeneity of the tail diagnostic; a regular-vine copula with edge-specific families would. We do not fit one, for a principled reason: the comonotone arm realizes the supremum over *all* copulas, vines included, so it upper-bounds whatever any heterogeneous-dependence model could achieve. If the maximal-coupling arm recovers no material value, no vine can. The sample-average CVaR program (9) is solved over the *same* feasible set $\mathcal{X}$ (4) and regimes as Phase 1, so the only thing that varies across arms is the dependence model; each schedule is evaluated once on the 2025 panel by out-of-sample $\mathrm{CVaR}_{0.95}$. Algorithm 3 summarizes the procedure.

The copula captures the asymmetry the covariance cannot. Figure 20 confirms the Clayton copula fitted to each grid behaves as intended, its samples pile into the joint-clean corner ($\chi_{\mathrm{L}} = 0.70$ vs. $\chi_{\mathrm{U}} = 0.26$ for CISO–BANC), exactly the structure the symmetric Gaussian copula ($\chi_{\mathrm{L}} \approx \chi_{\mathrm{U}} \approx 0.46$) and any covariance ball cannot represent.

Yet no copula recovers material value. When each dependence model is used to schedule and evaluated out-of-sample on 2025 (Table 11, Figure 21), no fitted copula beats the independence

Algorithm 3 Copula-scenario CVaR scheduling (one region set)

Input: daily panels $\rho_{1:N}$ (train 2021–2024, test 2025); dependence models {independence, Gaussian, Clayton, comonotone}; regimes; α-levels; scenarios $S = 1000$, fixed seed
for each dependence model C **do**
    fit C on training daily summaries (rank pseudo-observations); for Clayton, $\theta = 2\tau/(1-\tau)$ from mean Kendall τ
    for each regime, each α **do**
        draw S copula samples $u^s \in [0,1]^R$ ▷ indep./Gaussian/Clayton frailty/comonotone
        map each region's u^s_r to the training day of that daily-summary rank; take its full profile $\Rightarrow$ scenario $\rho^s \in \mathbb{R}^{R\times T}$
        solve the CVaR-SAA program (9) over $\mathcal{X} \Rightarrow$ schedule x
        evaluate *once* on 2025 $\Rightarrow$ out-of-sample $\mathrm{CVaR}_{0.95}$
copula gaps vs. independence + paired day-bootstrap CIs; comonotone realizes $\sup_C$, the ceiling Λ

baseline by a material, sign-stable margin; the Gaussian and Clayton gains never exceed 0.16% of $\mathrm{CVaR}_{0.95}$ and flip sign across α exactly as in Phase 1. To test the *actual* worst case we add the comonotone (upper-Fréchet) arm: perfect rank coupling. Since CVaR is comonotone-additive and the upper Fréchet bound is the most positively-associated coupling consistent with the marginals, comonotone maximizes the CVaR of the resulting sum over all copulas [7, 16], so it realizes the supremum $\sup_C$ in the leverage Λ of Section 3.10. At the locked scenario seed its largest single-cell gain is +0.18% (Eastern US–Canada), and a naive day-bootstrap would call several Eastern US–Canada cells "detectable." That detectability is illusory: the bootstrap resamples test days but not the *scenario draw*, and re-running the comonotone arm across five scenario seeds shows the gap is sampling noise, ranging from −0.08% to +0.10% and centred on zero (mean +0.01%; Figure 19). Even the maximal copula's leverage Λ sits at the scenario-sampling *noise floor*, $|\Lambda| \lesssim 0.1\%$ of $\mathrm{CVaR}_{0.95}$ and statistically indistinguishable from zero. This measurement is *independent* of the mean-leveling test, so it pins Λ without circularity: no copula, not even the maximal one, recovers material value.

One positive signal is worth stating precisely: the Clayton−Gaussian increment is consistently positive in the heterogeneous case (Diversified, mean +0.07%) and negligible or negative where correlation is common-mode. The asymmetric copula therefore *does* extract something the elliptical one cannot, in the expected direction, but at a magnitude an order below materiality. This is the empirical face of Proposition 1: the copula's leverage Λ is real but dominated by the mean field. The clearest statement of the result is out of sample (Figure 22): the distributions of real 2025 daily emissions under the independence, Clayton, and comonotone schedules lie on top of one another, with $\mathrm{CVaR}_{0.95}$ values agreeing to the third significant figure. The screening rule survives the richest passive dependence model available, exactly as Proposition 1 predicts: the binding constraint is the

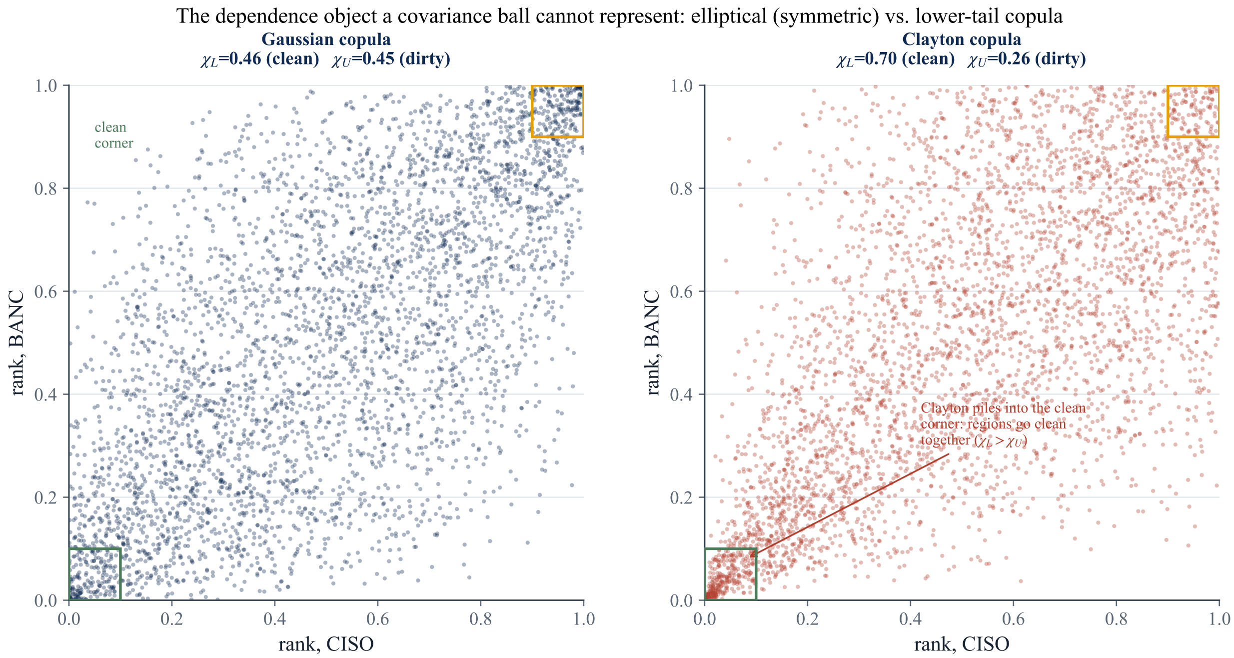

Figure 20: The dependence object a covariance ball cannot represent. Samples from the Gaussian copula (left, symmetric: $\chi_{\mathrm{L}} \approx \chi_{\mathrm{U}}$) and the fitted Clayton copula (right) for a representative US-West pair. Clayton concentrates in the lower-left "clean" corner ($\chi_{\mathrm{L}} > \chi_{\mathrm{U}}$), reproducing the Phase 1 radial asymmetry that elliptical models force to zero.

mean field, not the shape of the dependence object.

C The Five-Part Research Programme: Status and Horizon

This extended thesis gives a first, end-to-end treatment of all five parts of the programme; this appendix records what is settled versus what each part's full version still demands. **Part 1** (covariance null and mean-dominance bound, Sections 4–5) and **Part 2** (the copula extension, Section 4.8) are at full rigour: pre-registered, robustness-checked, equivalence-tested. **Part 3** (the active transfer DRO and the tail-risk crossover, Section 6) is bootstrap-quantified and code-validated; its remaining step is a *calibrated* emergency model and a pre-registered protocol, ideally with affine recourse rules. **Part 4** (the online rolling-horizon controller, Section 7) demonstrates the closed-loop result; its full version is a true multistage controller with a *learned* forecast (learning-augmented DRO), uncertain demand and transmission, and evaluation on real operational traces, the deployable-systems contribution. **Part 5** (the problem-class condition, Section 8) supplies the mean-dominance ratio Δ and validates it as a monotone regime indicator; its full version tightens the bound B (so the screen becomes conclusive, not merely a negative filter) and calibrates an absolute threshold, then tests the condition on other allocation domains, energy storage, EV fleets, spatial supply chains. In short,

Table 11: Copula results. Gain = reduction in out-of-sample $\mathrm{CVaR}_{0.95}$ versus the independence baseline (%; positive = the copula helps), the maximum *positive* gain over the nine cells at the locked scenario seed. The *comonotone* arm is the upper Fréchet bound (perfect rank coupling), the supremum over all copulas, hence the empirical ceiling Λ. The final column reports the comonotone gap re-evaluated over five scenario seeds: in every case it is centred on zero and sampling-noise dominated, so no gain is stable or material.

Case	$\bar{\tau}$	θ	$\mathrm{gain}_{\mathrm{Gauss}}$	$\mathrm{gain}_{\mathrm{Clay}}$	$\mathrm{gain}_{\mathrm{comon}}$	comon. over 5 seeds
Western US	0.47	1.77	0.066	0.038	0.089	≈ 0, noise
Eastern US–Canada	0.31	0.90	0.158	0.014	0.182	[−0.08, +0.10], mean +0.01
Diversified	0.20	0.50	0.069	0.127	0.114	≈ 0, noise

the arc is complete in outline and rigorous at its core (Parts 1–2); Parts 3–5 are validated first cuts whose sharpening is the post-capstone research agenda, detailed in the project's research roadmap.

Annex A: Individual Contribution Statement

This Research Capstone was completed individually by **Marco Ortiz Togashi** under the supervision of Prof. Bissan Ghaddar. As sole author, I am responsible for all components of the work:

- **Research design.** Formulated the three research questions, designed the shuffled-marginals falsification test, and specified the pre-registration protocol that separates assumption validity from value.
- **Modelling.** Specified the Mahalanobis–Wasserstein DRO scheduling model and its SOCP reformulation, the constraint regimes (including the variable carbon-coupled capacity constraint adapted from the SoCC'25 formulation), and the mean-leveling test and tail-dependence diagnostics.
- **Data and implementation.** Assembled the multi-grid carbon-intensity and temperature panels, implemented the full experimental pipeline, the robustness battery, the Benjamini–Hochberg correction, and the figures, under version control with a continuous-integration test suite.
- **Analysis and writing.** Produced and interpreted all results, established the mechanistic explanation, and wrote the manuscript in full.

Generative AI tools were used as a coding and drafting assistant under my direction, as declared on the cover page; all research questions, modelling decisions, experimental designs, interpretations, and final text are my own. The inter-region migration mechanism itself follows established carbon-aware practice (Section 2); my individual contribution, developed under the supervisor's direction,

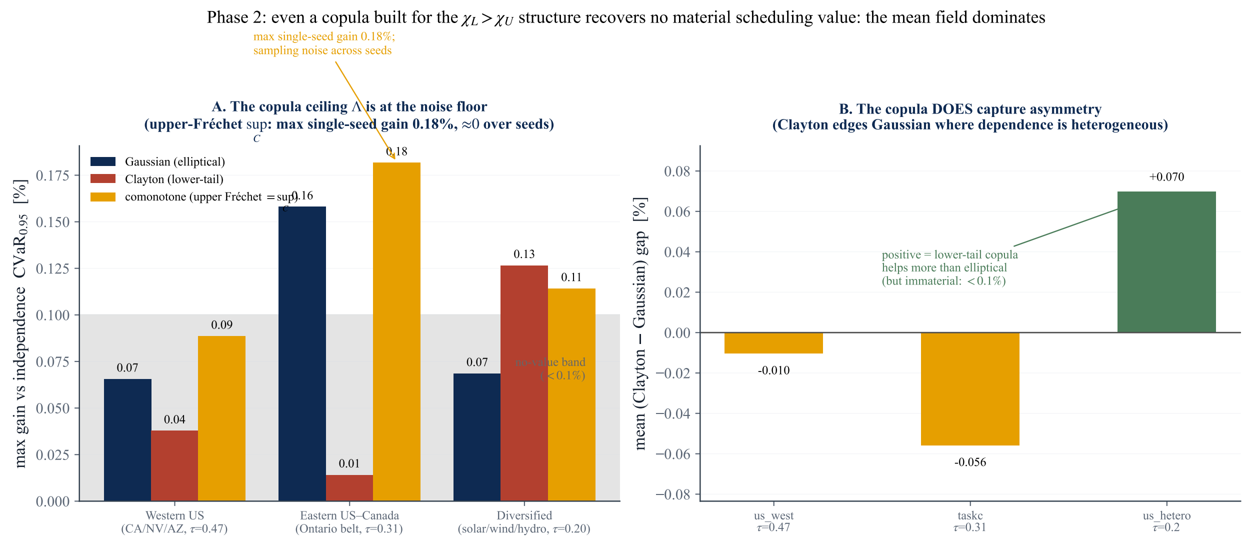

Figure 21: Copula result (single locked scenario seed). **A:** maximum positive gain versus independence for the Gaussian, Clayton, and comonotone (upper-Fréchet $\sup_C$) arms. The largest single-seed comonotone gain is 0.18% (Eastern US–Canada); re-evaluated over scenario seeds it is sampling noise centred on zero (Figure 19). **B:** the mean Clayton−Gaussian increment by grid; positive where dependence is heterogeneous (Diversified, +0.07%), confirming the copula *does* capture the asymmetry, but an order of magnitude too small to matter, as Proposition 1 predicts.

is its distributionally robust treatment, the honest $\Phi = 0$-anchored measurement of its spatial value, and the screening and price-of-robustness decision rules.

Marco Ortiz Togashi ____________________ Date ____________________

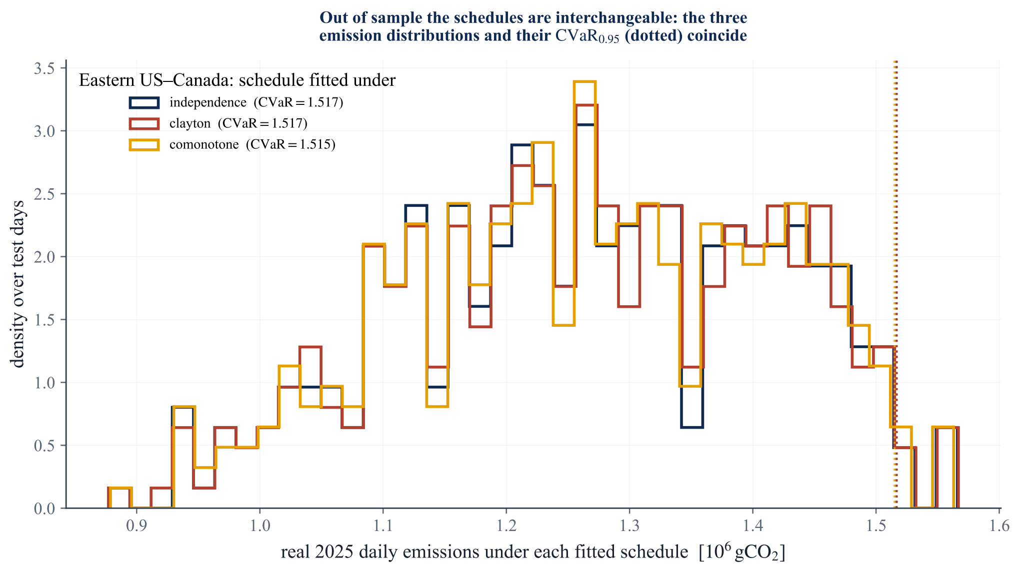

Figure 22: Out of sample the copula schedulers are interchangeable (Eastern US–Canada). Distribution of real 2025 daily emissions under the schedule fitted to independence, Clayton, and comonotone scenarios; the dotted lines mark each $\mathrm{CVaR}_{0.95}$. All three coincide.